Department of Electrical and Computer Engineering

North South University

Senior Design Project

A Voice-Based AI Framework for Medical Pre-Screening of Bangla and Regional Bangladeshi Speech

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Summer, 2026

LETTER OF TRANSMITTAL

September, 2026

To
Dr. Mohammad Abdul Matin [mtn]
Professor & Chair
Department of Electrical and Computer Engineering
North South University, Dhaka

Subject: Submission of Capstone Project Report on "A Voice-Based AI Framework for Medical Pre-Screening of Bangla and Regional Bangladeshi Speech".

Dear Sir,

With due respect, we would like to submit our Capstone Project Report on "A Voice-Based AI Framework for Medical Pre-Screening of Bangla and Regional Bangladeshi Speech" as a part of our B.Sc. program in Computer Science and Engineering. The report covers the work of both semesters of the Senior Design Project. In CSE499A we assembled a multi-dialect Bangla speech corpus and benchmarked eighteen open-source speech recognition and multimodal audio models on it. In CSE499B we used that measurement to design and build a working pre-screening system with three web portals, a conversational follow-up loop, a rule-based safety check, and standards-based clinical output.

We have tried to follow the BAETE Capstone Final Report template, the guidelines of the department, and the requirements set by our faculty advisor. Throughout the report we have been careful to separate what we measured from what we built and from what remains planned, and we have stated the limits of our evaluation openly rather than leaving them out.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative for the assessment of our Senior Design work.

Sincerely Yours,

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APPROVAL

Rafiur Rahman Mashrafi (ID# 2221971042), M. G. Rabbi Hossen (ID# 2222516042) and Israt Zaman Srity (ID# 2211084042) from the Department of Electrical and Computer Engineering, North South University, have worked on the Senior Design Project titled "A Voice-Based AI Framework for Medical Pre-Screening of Bangla and Regional Bangladeshi Speech" under the supervision of Dr. Mohammad Ashrafuzzaman Khan [AzK] in partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering and have submitted it satisfactorily.

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

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ACKNOWLEDGEMENTS

The authors would like to express their heartfelt gratitude towards their project and research supervisor, Dr. Mohammad Ashrafuzzaman Khan (AzK), Associate Professor, Department of Electrical and Computer Engineering, North South University, Bangladesh, for his constant guidance, technical mentorship and honest feedback across both semesters of this Senior Design Project. His insistence that the team measure the speech recognition problem before designing around it changed the direction of the whole project, and his review of the system design helped us keep the safety rules at the centre of the work rather than at the edge of it.

The authors would also like to thank the Department of Electrical and Computer Engineering, North South University, Bangladesh, for providing the academic environment and the laboratory facilities in which this work was carried out. Finally, the authors are grateful to the open-source community, whose freely published speech models, libraries and tools made it possible to complete a project of this size without any specialised hardware.

ABSTRACT

A Voice-Based AI Framework for Medical Pre-Screening of Bangla and Regional Bangladeshi Speech

Outpatient consultations in Bangladesh are short, and a large part of that short time is spent collecting the patient history by hand. Patients who speak a regional dialect or mix Bangla with English are served poorly by digital health tools that expect typed, standard language. This Senior Design Project set out to let a patient describe the problem by speaking naturally before the consultation begins, and to turn that conversation into a structured record the doctor can read in a few seconds. The project was carried out in two stages. The first stage measured the hardest part of the problem. A multi-dialect Bangla speech corpus of about four and three-quarter hours, covering five regional varieties, was assembled and preprocessed, and eighteen open-source speech recognition and multimodal audio models were evaluated on it using Word Error Rate, Character Error Rate and a translation quality score. The strongest model still misread close to half of the words, and every one of the larger models performed worse than the smaller Bangla-specialised models, with two of them failing to produce usable output at all. The second stage used that result as a design constraint rather than a discouragement. Because the transcription cannot be trusted, the system was built to preserve the patient's exact words permanently, to clean and structure them only in later and separately stored steps, to close the remaining gaps by asking spoken follow-up questions, and to keep the emergency safety check as a local rule that no language model can suppress. The delivered system has a patient kiosk, a triage desk portal and a doctor portal over one backend service with an eighteen table database, and it exports each encounter as an editable document, a standard health interchange record and a printable summary. The behaviour of the system is protected by a large automated test suite, and one supervised session with a real microphone confirmed that the full voice loop works end to end. The system itself has not yet been evaluated with formal accuracy or timing measurements, and this report states plainly which claims rest on measurement and which rest only on the tests and that single supervised session.

Keywords: Bangla speech recognition, dialect speech, code-mixed speech, electronic health records, clinical pre-screening, low-resource language processing, voice user interface, multimodal audio models.

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Chapter 1 Introduction

1.1 Background and Motivation

Bangladesh has far fewer doctors than its population needs. World Bank figures, compiled from national health workforce reporting, put the country at about 0.64 physicians per 1,000 people [1], which is fewer than one doctor for every thousand patients. In practice this means outpatient departments in public hospitals and in most private chambers work through long queues, and each patient gets only a few minutes with the physician.

A noticeable part of those few minutes is not spent on examination or on clinical reasoning at all. It is spent on history taking: asking the patient what the problem is, when it started, what it feels like, what medicines they are already on, and whether they have any allergy. These questions are the same for almost every patient, they are asked verbally, and the answers are usually written by hand on a prescription pad that nobody digitises afterwards. The information is collected once and then lost, so a patient who returns three weeks later is asked the same questions again.

The problem is harder in Bangladesh than the description above suggests, because of how patients actually speak. A patient from Sylhet, from Barishal or from the older parts of Dhaka does not speak the standard Bangla of textbooks and news broadcasts. Many patients also mix English words into Bangla sentences, a habit usually called Banglish, and this happens especially with medicine names, body parts and time expressions. On top of this, elderly and low-literacy patients often cannot type at all, which rules out most existing digital intake forms. Any system that expects clean typed standard Bangla will therefore fail exactly the patients who would benefit from it most.

Speech technology is the obvious answer, but Bangla is a low-resource language for speech recognition. Although it has a very large number of speakers, the amount of publicly available labelled Bangla audio is small compared with English, and almost all of it is standard Bangla rather than dialect speech. Medicine names and clinical vocabulary are rare words that recognisers have mostly not seen, and a clinic waiting area is a noisy place. The practical question at the start of this project was therefore not whether voice input would be convenient, but whether Bangla speech recognition is accurate enough to be safe in a medical setting at all.

That question shaped the entire project. Rather than assume that transcription would be good enough, the team measured it first, and then designed the system around the answer.

1.2 Purpose and Goal of the Project

The goal of this project is to build a voice-based pre-screening system for Bangladeshi clinics. The patient speaks naturally in Bangla, in Banglish or in a regional dialect before meeting the doctor. The system captures what they said without changing it, cleans and structures the text in later steps, asks spoken follow-up questions to fill in whatever is still missing, assesses how urgent the case looks, explains that assessment in plain language, and hands the doctor a structured record before the consultation begins.

Four requirements were fixed at the start of the project and were used to judge every design decision afterwards.

R1. Real-time processing. Speech should be transcribed while the patient is speaking, with little delay between speaking and seeing text.

R2. Fidelity to the spoken words. The displayed text should match what the patient actually said. Words must not be silently replaced or reworded.

R3. Bangladesh-focused recognition. The system must handle standard Bangla, regional dialects and Bangla–English code-switching, across different ages, speaking speeds and educational backgrounds.

R4. Open-source and low-cost deployment. Open-source models and tools are preferred, the system must run on ordinary computers without a dedicated graphics processor, and any external service must sit behind a swappable interface.

Alongside these requirements, four rules were treated as non-negotiable throughout the work, because a pre-screening tool that breaks any of them is unsafe rather than merely inconvenient. The raw transcript is never edited. The system never gives a diagnosis. Danger signs are always surfaced and the patient is never falsely reassured. Only synthetic or consented data is used during development. Chapter 3 describes how each of these rules is enforced in the code itself rather than left to discipline.

The work of the project falls into three groups, and this report keeps them separate on purpose.

What was measured. A five-variety Bangla speech corpus was collected and prepared, and eighteen open-source speech models were benchmarked on it. The results, given in Chapter 4, are the only quantitative results the project has, and they are the clearest research contribution of the work.

What was built. A complete pre-screening system with three web portals, a sixteen-module processing pipeline, an eighteen-table database, a rule-based emergency safety check, and clinical output in an editable document format, in a standard health interchange format and as a printable summary.

What is proposed. Fine-tuning a mid-sized Bangla speech model, replacing the cloud recogniser with a local one, adding real authentication and encryption, and running a formal evaluation of the built system. None of these were completed, and none are described in this report as if they were.

1.3 Organization of the Report

The rest of this report is arranged as follows. Chapter 2 reviews the research literature on multilingual and Bangla speech recognition, on large multimodal audio models, and on clinical language processing, and states the gap this project addresses. Chapter 3 describes the methodology: the system design, the software components chosen and why, and how the system was implemented. Chapter 4 presents the speech corpus, the two benchmark experiments, the analysis by dialect, the verification carried out on the built system, and an explicit statement of what was not measured. Chapter 5 discusses the impact of the project on society, health, safety, legal and cultural issues, and on the environment and sustainability. Chapter 6 gives the project timeline and budget. Chapter 7 maps the work against the complex engineering problem and complex engineering activity attributes. Chapter 8 summarises the project, states its limitations honestly, and describes the future work that follows from them.

Chapter 2 Research Literature Review

2.1 Existing Research and Limitations

This project draws on four separate research areas: large multilingual speech recognition, the newer multimodal audio language models, Bangla speech resources, and clinical language processing. Each area is reviewed below, and each review ends with what that line of work does not yet solve for a Bangladeshi clinic.

2.1.1 Multilingual and Large-Scale Speech Recognition

Almost all modern speech recognition is built on the Transformer architecture [2]. Radford et al. [3] trained *Whisper*, an encoder–decoder model, on roughly 680,000 hours of weakly labelled multilingual audio collected from the web, and showed that it transcribes close to a hundred languages without any language-specific training. Pratap et al. [4] pushed coverage much further with *Massively Multilingual Speech*, which reaches over 1,100 languages by combining self-supervised pre-training with connectionist temporal classification fine-tuning on religious-text recordings. A different route is self-supervised representation learning: Baevski et al. [5] proposed *wav2vec 2.0*, which learns speech representations from unlabelled audio before any transcription labels are used, and Conneau et al. [6] extended the same idea across languages as XLSR-53. Many of the open Bangla recognisers available today are fine-tuned versions of these two families. The Seamless Communication team's *SeamlessM4T* [7] takes yet another approach and handles speech and text translation in one model.

These are strong general systems, but none of the papers above reports results on dialectal Bangla, and none reports results on medical speech in any Indic language. Their published Bangla numbers come from read-aloud corpora recorded in quiet conditions, which is not the situation this project is concerned with.

2.1.2 Multimodal Audio Language Models

A more recent direction attaches a speech encoder to a language-model decoder, so that one model both hears and reasons. Chu et al. [8] described the *Qwen2-Audio* family, which accepts audio together with a text instruction, and the same group later released a speech-recognition variant, Qwen3-ASR. Mistral AI's *Voxtral* [9] and Microsoft's *Phi-4 Multimodal* [10] follow the same pattern at different model sizes.

The common assumption around these models is that a larger model with broader training will also be better on languages that were poorly represented in that training. That assumption has been tested very little on dialect-heavy low-resource speech. Testing it directly on Bangladeshi dialect audio is one of the contributions of this project, and the result reported in Chapter 4 does not support the assumption.

2.1.3 Bangla Speech Resources

Several public Bangla speech resources exist. The Bengali.AI speech recognition competition [11] produced a read-aloud Bengali benchmark and motivated a number of the fine-tuned Whisper and wav2vec 2.0 checkpoints that this project evaluated. Mozilla's *Common Voice* [12] provides a crowdsourced Bengali corpus recorded by volunteers. On the Indic side, AI4Bharat's *IndicWav2Vec* [13] extends self-supervised speech models to twenty-two Indian languages, the *Vakyansh* project [14] publishes recognisers tuned for Indic speech including Bengali, and *Shrutilipi* [15] mines paired audio and text from public broadcasts to build one of the largest open Indic speech datasets.

The work closest to this project is *OOD-Speech* by Rakib et al. [16], a Bengali corpus built specifically to test how recognisers behave on speech that differs from their training distribution. That paper varies speaker, recording condition and register. This project is concerned with a narrower but harder combination: regional dialect, Bangla–English code-mixing, and informal clinical description, all at the same time. We did not find a published benchmark for that intersection.

2.1.4 Clinical Language Processing and Voice in Medicine

In the clinical domain, domain-adapted language models have repeatedly been shown to beat general ones. Lee et al. [19] pre-trained *BioBERT* on biomedical literature, and Alsentzer et al. [20] pre-trained *ClinicalBERT* on hospital notes; both improved downstream clinical extraction tasks over a general BERT model [18]. For Bangla, Bhattacharjee et al. [17] released *BanglaBERT* with benchmarks for several Bangla language tasks. There is, however, no published Bangla equivalent of BioBERT or ClinicalBERT, no public Bangla medical named-entity dataset, and no Bangla clinical extraction benchmark. This absence is the reason the present system extracts clinical fields using instruction-following language models behind a swappable interface, rather than a fine-tuned

Bangla clinical model, which would first have required a labelled Bangla medical corpus that does not exist.

On the product side, the closest commercial systems are the ambient clinical scribes that listen to a doctor–patient conversation and draft a note. These systems are English-only, are deployed in well-resourced hospital systems, and integrate with established electronic health record platforms. They also assume the *doctor* is the speaker. Older voice systems for medical dictation make the same assumption. This project reverses it: the *patient* speaks, before the consultation rather than after it, and the artefact produced is the intake record the doctor reads on the way in.

2.1.5 Efficient Adaptation of Large Models

Fine-tuning a large speech model on a small dialect corpus is expensive if every weight is updated. Hu et al. [21] introduced *Low-Rank Adaptation*, which freezes the original weights and trains only small adapter matrices, cutting the compute and memory cost of adaptation dramatically. This is the method the project has planned for future fine-tuning work. No fine-tuning was performed during either semester, and Section 8.3 describes it as future work rather than as a result.

2.2 Limitations of Existing Work

Reading across the four areas above, four gaps stand out, and together they define what this project attempted.

First, dialectal Bangla clinical speech has not been benchmarked. General multilingual recognisers report Bangla numbers on clean read speech. Bangla-specific resources are mostly standard Bangla. No published work measures what happens when regional dialect, code-mixing and informal medical description occur together, which is exactly what a Bangladeshi clinic hears.

Second, systems are usually designed as if transcription were reliable. Most voice-to-record pipelines treat the transcript as an input to trust. If the recogniser is wrong roughly half the time, as this project measured, a single-pass design silently loses clinical information and nobody notices.

Third, there is no conversational mechanism for closing the gaps. Existing work transcribes and extracts, but does not check what is still missing from the clinical picture and then ask the patient about it in their own language.

Fourth, the deployment assumptions do not fit a small Bangladeshi clinic. The commercial systems assume English, reliable connectivity, an existing electronic record system to integrate with, and a budget. A usable system here has to work in Bangla, run on ordinary computers without a dedicated graphics processor, and cost nothing to try.

This project addresses the first gap by measuring, and the second, third and fourth by designing the system around the measurement rather than around an assumption.

Chapter 3 Methodology

3.1 System Design

3.1.1 Design Constraints

Three constraints were fixed before any code was written, and they explain most of the decisions in this chapter.

The first is the hardware. The team had no machine with an NVIDIA graphics processor. Development was done on a desktop with a six-core AMD processor and 24 GB of memory and on a laptop with a six-core AMD processor and 12 GB of memory, and the system had to run identically on Windows and on Linux from a single requirements file. Anything that needed a training-capable graphics card was therefore out of reach, and any component that assumed one was rejected.

The second is cost. The project rule was free or open-source first. Paid application programming interface tiers were not used at any point, so every external service in the system is a free tier and every model is an openly licensed checkpoint.

The third is the accuracy of Bangla speech recognition. The benchmark reported in Chapter 4 showed that the best open model still misread close to half the words of dialect speech. A system that assumes its transcript is correct would therefore lose clinical information silently. This single measurement is the reason the design looks the way it does.

3.1.2 The Four Rules

Four rules were treated as invariants for the whole project. They are stated here because almost every later design choice is a consequence of one of them.

Rule 1 — the patient's exact words are never changed. What the recogniser produced is stored once and never updated. Cleaning and correction happen in a later stage and are written to a separate field. No portal offers an edit control over the raw text.

Rule 2 — the system never diagnoses. It narrows the search space for the doctor. Where the system does surface a possible condition for staff, it is labelled as a suggestion, carries a disclaimer attached on the server side, and is excluded from the exported clinical record.

Rule 3 — danger signs are surfaced and the patient is never falsely reassured. A separate emergency detection module was removed early in the design and its responsibility was moved into the risk assessment stage as a local rule, so that the check survives any failure of the language models.

Rule 4 — patient data is sensitive. Free model providers may train on what is sent to them, and the browser speech recogniser sends audio to a cloud service, so only synthetic or consented sample data was used throughout development. No real patient data was processed at any point.

3.1.3 Overall Architecture

The system is a single backend service with three separate web front ends, shown in Figure 1. All three front ends are plain HTML, CSS and JavaScript served as static files by the same application, so there is no separate build step and no JavaScript framework to install.

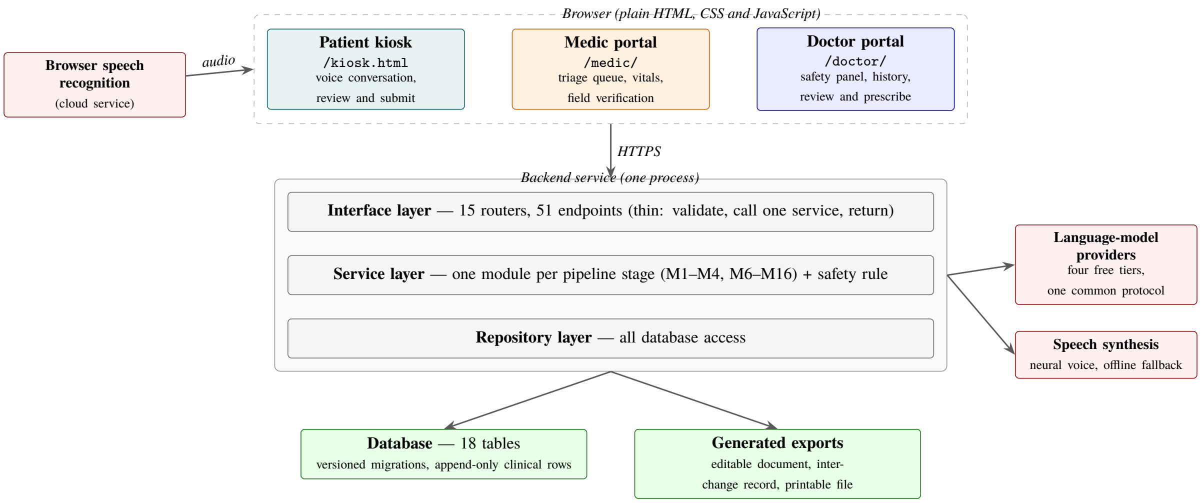

Figure 1. Architecture of the pre-screening system. One backend service serves three separate portals and mediates every call to an external speech or language service.

The backend exposes fifty-one endpoints across fifteen router modules, plus a health check. Routers are deliberately thin: they validate the request, call one service function and return a response model. The clinical logic lives in the service layer, with roughly one module per pipeline stage, and all database access goes through repository modules. This layering means that adding a capability usually touches one file per layer instead of spreading across the codebase.

External services are reached only from the backend, never from the browser, with one deliberate exception: speech recognition runs in the patient's browser because the browser interface is the only free real-time Bangla recogniser available to the project.

3.1.4 The Processing Pipeline

Work is organised as a numbered set of modules. Figure 2 shows the path a single patient's visit takes through them.

Module 5 was originally a standalone emergency detection stage that raised an alert to staff. It was removed during the design review, and its safety job moved inside Module 10 as a deterministic rule. The number 5 was left as a permanent gap rather than renumbering the later modules, because the project's decision records already referenced modules 6 to 15 by number. Module 16 was added later as a doctor-facing information assistant.

Table I lists every module with its current status. The statuses are taken from the project's own status board and were checked against the source tree while writing this report.

TABLE I. Modules of the system and their implementation status.

#	Module	Status	What it does
M1	Speech-to-Text	Implemented	Captures the patient's speech in the browser and stores the raw text unchanged.
M2	Text Correction	Implemented	Fixes spelling and grammar into a separate field; never rewrites the raw text.
M3	Information Extraction	Implemented	Fills ten fixed clinical fields, each in English and Bangla.
M4	Initial Summary	Implemented	Produces a short chief-complaint summary from the extracted fields.
M5	Emergency Detection	Retired	Removed as a stage; its job is now the local red-flag rule inside M10.
M6	Missing Information	Implemented	Compares the extracted fields against what a case needs and reports the gaps.
M7	Follow-up Questions	Implemented	Generates the next question, shown on screen and spoken aloud.
M8	Response Processing	Implemented	Merges the answer into the profile without overwriting staff corrections.
M9	Completion Check	Implemented	Scores completeness locally and decides whether to loop again.
M10	Risk Assessment	Implemented	Assigns a risk tier; a local rule forces the highest tier on danger signs.
M11	Explainable Output	Implemented	Writes a plain-language reason for every risk assessment.
M12	Structured Report	Implemented	Assembles the doctor-facing report locally, with no diagnosis.
M13	Record Storage	Implemented	Stores the encounter across eighteen tables with an append-only audit trail.
M14	Doctor Dashboard	Implemented	Presents the case, safety panel, history and review controls.

continued on next page

Table I continued from previous page

#	Module	Status	What it does
M15	Feedback Loop	Partial	Doctor feedback is captured and stored; the retraining pipeline is not built.
M16	Drug/Test Assistant	Implemented	Answers a doctor's medicine or test question with a server-attached disclaimer.

3.1.5 Roles and Data Ownership

The three portals are three roles, not three views of the same screen. Each one answers a different question and owns a different part of the record.

The **patient kiosk** answers "tell us what is wrong, in your own words." It is the only place where the raw transcript is created. It never shows a queue, a risk tier or a suggested condition.

The **medic portal** is the triage desk. It answers "who do I handle next, and is this case fit to hand to a doctor?" It orders the waiting queue by urgency rather than by arrival time, records the patient's identity and vital signs, verifies the automatically extracted fields against the patient's own words, and forwards the case to a named doctor.

The **doctor portal** answers "what is going on with this person, and what do I do about it?" It shows the safety information first, then the patient's words, the extracted fields, the previous visits and prescriptions, and the controls for reviewing the case and writing a prescription.

The handover between them is one-directional and is carried entirely by the visit's status field: `in_progress` when the patient is speaking, `awaiting_review` once they submit, `awaiting_doctor` once the medic forwards, and `reviewed` once the doctor is finished. There is no message passing and no second copy of the case; the medic and the doctor read and write the same rows.

3.1.6 Database Design

The schema has eighteen tables and is managed by fourteen versioned migrations that run automatically when the application starts. Figure 3 shows the main relationships.

Four principles shaped the schema.

The visit is the aggregate root. One pre-screening encounter is one visit row, and every output of the pipeline hangs off it. Adding a new kind of output later means adding a child table, not reshaping an existing one.

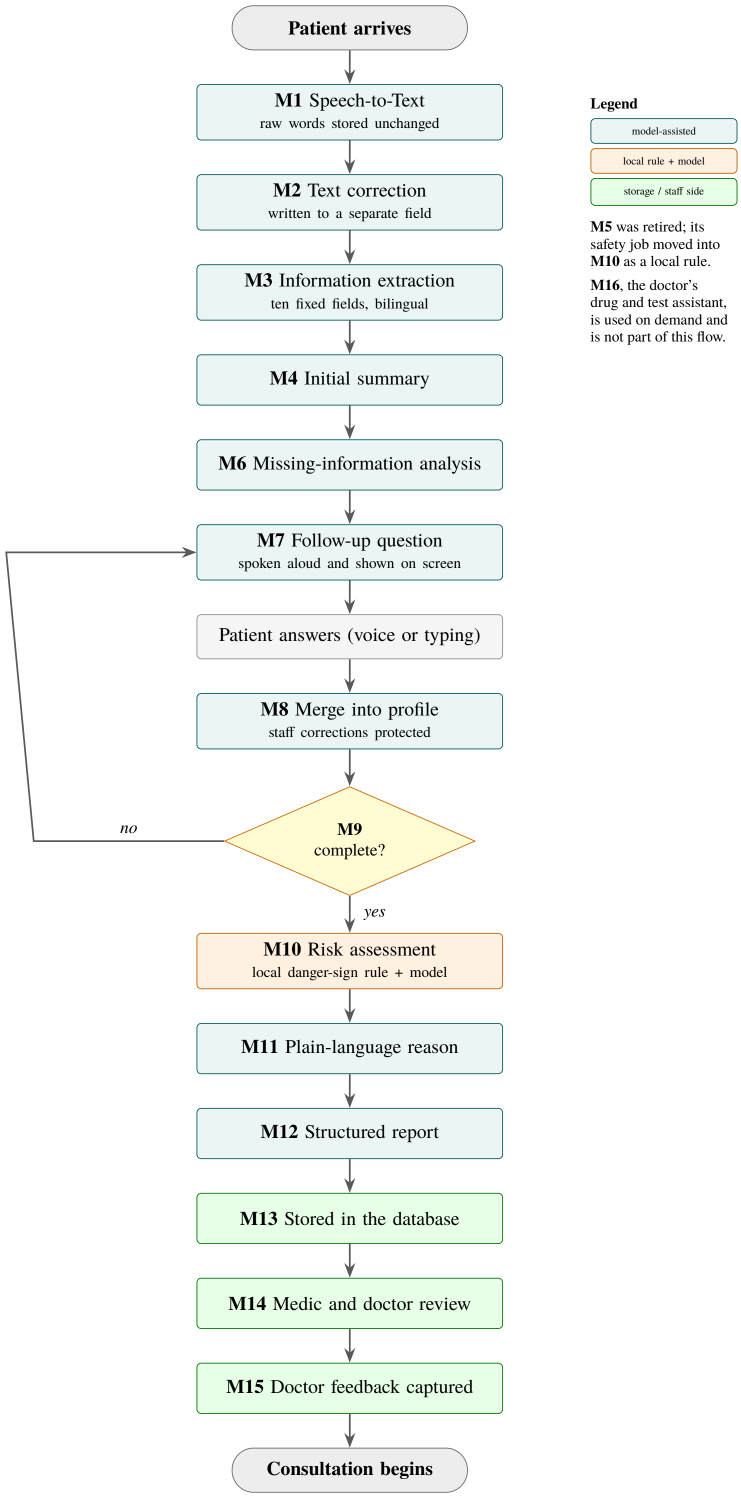

Figure 2. Journey of one patient visit through the processing modules. The decision at Module 9 loops back to Module 7 until the case is complete enough or the question budget is spent.

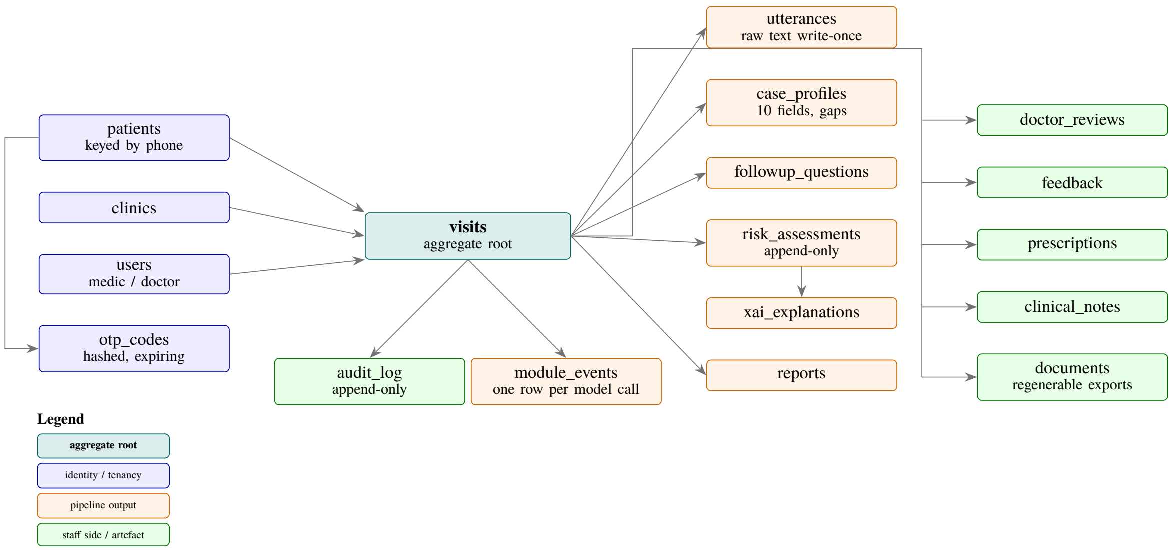

Figure 3. Main entities of the database. The visit is the aggregate root; every output of the pipeline is a child row that points back to it.

Clinical and accountability rows are append-only. Utterances, risk assessments, reviews, feedback and audit entries are never updated in place. A new state is a new row. This preserves the history of a case and is what makes Rule 1 enforceable rather than merely intended.

Evolving structured data lives in JSON columns. The extracted fields, the information gaps, the risk drivers and the report sections are stored as JSON rather than as columns, so that adding a new field is a data change and not a schema migration. A field is promoted to a real column only when the system needs to filter or sort on it.

Module runs are logged as data. Every model call is recorded in a `module_events` row with its module code and the provider that served it. A future module needs no schema change, and the record doubles as the project's own observability data.

Table II lists the tables and what each one owns.

TABLE II. Database tables and the data each one owns.

Table	Owns
`clinics`	The clinic, and the letterhead used on printed documents.
`users`	Staff accounts and their role (medic, doctor, admin).
`patients`	The person, keyed by phone number: name, sex, age, height, weight, blood pressure and blood glucose.
`visits`	One pre-screening encounter and its status through the handover chain.

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Table II continued from previous page

Table	Owns
`utterances`	Every spoken or typed turn. The raw text column is written once and never updated.
`case_profiles`	The derived working state of a visit: extracted fields, summary, gaps and completeness score.
`followup_questions`	Each generated question and the answer it maps to.
`risk_assessments`	Append-only risk rows, the triggered danger signs, and whether the rule overrode the model.
`xai_explanations`	The plain-language reason for one risk assessment.
`reports`	The assembled doctor-facing report sections.
`doctor_reviews`	The doctor's decision and notes on a visit.
`feedback`	Doctor feedback captured for future model improvement.
`prescriptions`	Prescriptions written by a doctor, with the medicines, advice and required tests.
`clinical_notes`	Recall reminders and messages addressed to a role rather than a person.
`documents`	Generated export files; regenerable, so the database stays the source of truth.
`module_events`	One row per module run, with the provider that served it.
`otp_codes`	Hashed one-time codes with their expiry and attempt count.
`audit_log`	Append-only record of who changed what, including values written by the system itself.

3.2 Software Components

This is a software-only project. No hardware was designed or fabricated. In a real deployment the patient-facing device would be an ordinary tablet or a touchscreen computer in the waiting area, running a web browser.

Table III lists the tools actually used in the delivered system, together with the alternatives that were considered and the reason for the choice. The reasons are drawn from the project's own decision records, of which sixty-nine were written during the two semesters.

TABLE III. Software tools and components used in the project.

Tool	Function	Other similar tools	Why selected this tool
Python	Backend language	Java, Node.js	One language for the web service and for the evaluation notebooks; runs unchanged on Windows and Linux.

continued on next page

Table III continued from previous page

Tool	Function	Other similar tools	Why selected this tool
FastAPI [28]	Web framework and REST interface	Flask, Django	Request and response validation comes from type hints, and interactive documentation is generated automatically.
Uvicorn	Application server	Gunicorn, Hypercorn	Starts the whole system with one command, which suits a clinic with no system administrator.
SQLite [29]	Database	PostgreSQL, MySQL	No server to install; the whole database is one file. The connection string can be switched to PostgreSQL without a code change.
SQLAlchemy [30]	Database access	Django ORM, raw SQL	Tables are defined once as Python classes and work on both SQLite and PostgreSQL.
Alembic [31]	Schema migrations	Hand-written SQL scripts	Every schema change is a reviewable, reversible file, and the application migrates itself at startup.
Browser Web Speech interface [26]	Module 1 speech capture	faster-whisper, Vosk	The only free real-time Bangla recogniser available without a server or a key; local models need a graphics processor the team does not have.
OpenAI-compatible client	All language-model calls	One client per provider	Four different providers speak the same protocol, so switching provider is a configuration change.
Gemini, Groq, Cerebras, OpenRouter free tiers	The language-model providers	Paid model endpoints	Free, capable, and spread across independent daily quotas so no single limit stops the system.
edge-tts [35]	Spoken questions, server side	Google Cloud Text-to-Speech	Natural Bangla neural voice, free and keyless; used when the browser has no Bangla voice installed.
eSpeak NG [36]	Offline speech fallback	Festival, Flite	Fully local, so a clinic with no internet still hears the question.

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Table III continued from previous page

Tool	Function	Other similar tools	Why selected this tool
python-docx [32]	Editable document export	LibreOffice conversion	Writes the transcript, summary and prescription documents directly, with no office suite installed.
fpdf2 with HarfBuzz [33, 34]	Printable record	ReportLab	ReportLab cannot shape Bengali conjuncts. HarfBuzz can, and the renderer refuses to draw a character its font does not contain.
ddgs	Module 16 web search	Paid search APIs	Free and keyless; the doctor's typed question is the only thing sent.
httpx	SMS gateway calls	requests	Modern HTTP client already needed for the one-time-code channel.
pytest [37]	Automated testing	unittest	Concise test functions and fixtures; the suite is the main safety net against regressions.
Plain HTML, CSS and JavaScript	All three portals	React, Vue	No build step, no dependency tree, and the pages load quickly on a modest clinic machine.
Google Colab	Benchmark compute	Local GPU, Kaggle	Free graphics processor time, which the team's own machines do not have.
Hugging Face Transformers [22]	Running the benchmarked models	ESPnet, NeMo	One interface across every model family the benchmark covered.
WhisperX [23]	Audio segmentation for the benchmark	pyannote.audio	Splits long recordings at speech boundaries, which most of the evaluated models require.
jiwer and Hugging Face evaluate [25]	Error-rate and translation-quality scoring	NIST sclite	Pure Python, no external binary, and the standard choice for these metrics.
yt-dlp	Corpus collection	youtube-dl	Actively maintained; used to collect the publicly available dialect recordings.
Git and GitHub	Version control	GitLab, Bitbucket	Standard, and required for the reproducibility of the benchmark.

3.3 Software Implementation

3.3.1 Backend and Interface Surface

The backend is one application object that mounts fifteen routers and five sets of static files. The routers are registered before the static mounts so that interface calls always win over file serving. On startup the application runs any outstanding database migrations, fills in the demonstration clinic letterhead if it is empty, and writes a warning to the log for any model provider that has a key configured but no model named, because a single blank configuration line would otherwise remove a fallback provider silently.

Static files are served with a revalidation header. This was added after a real failure: the browser cached an old copy of a shared script, and the kiosk quietly fell back to the wrong speech language. On a clinic machine the same staleness would have disabled Bangla audio after an update, and nobody would have seen it happen.

3.3.2 Speech Input and Output

Module 1 uses the speech recognition interface built into the browser, set to Bangladeshi Bangla, in continuous mode with interim results enabled. Interim results are what make the patient's words appear on screen while they are still speaking. The recogniser runs in the patient's browser and sends audio to a cloud service, which is precisely why Rule 4 restricts development to synthetic data.

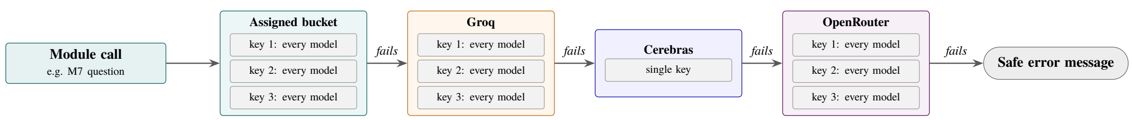
Speech output is layered. The browser's own speech synthesis is tried first because it needs no server and no key. If the machine has no Bangla voice installed, the kiosk falls back to a server endpoint that renders the question with a neural Bangla voice, and if that engine is missing as well, to a fully local offline voice. A missing engine returns an explicit error rather than a silent empty response, so a clinic never ends up with a kiosk that appears to be speaking but is not. Only the assistant's own question text is ever sent to this endpoint; no patient utterance passes through it.

The voice loop is deliberately hands-free but never traps the patient. The assistant speaks the question, the microphone opens itself after a short guard delay so the assistant does not hear its own voice, the patient answers, and a visible three-second countdown offers to close the turn. The countdown is a confirmation window and not a hard cut-off: if the patient starts speaking again, it is cancelled. A typing option is always visible next to the microphone, and both routes go through the same endpoint, differing only in a source field that records whether the answer arrived by microphone or by keyboard. The timings are served to the kiosk from the backend so that a clinic can tune them without editing JavaScript.

3.3.3 The Language-Model Layer

Nine of the sixteen modules call a language model. Rather than binding the system to one provider, every call goes through a single client that speaks the OpenAI-compatible protocol, and each module is assigned a provider bucket by configuration. Quality-sensitive tasks such as correction, summarisation, risk wording and explanation are assigned to a stronger model; cheap structured extraction is assigned to a lighter and higher-throughput model; the two tasks inside the live conversation loop are assigned to the fastest provider so the patient is not left waiting.

Because every provider is a free tier with a daily quota, the system needs to survive running out. Figure 4 shows the chain. Redundancy has three independent dimensions that multiply: the ordered list of provider buckets, up to four credential slots within a bucket, and one or more model names within a credential. Each attempt is one combination of the three. A rate-limit response puts only that exact combination to sleep for a short, fixed period, never the whole provider, and nothing about credential health is stored anywhere.



A rate-limit response puts to sleep only that one combination of bucket, credential and model, for a short fixed period. A whole provider is treated as failed only when every one of its credentials has failed. Nothing about credential health is stored, and no credential ever reaches a log line — only its slot number. If the entire chain fails, the patient is shown a fixed safe sentence; no provider name, model identifier or upstream error text ever reaches the screen.

Figure 4. The provider fallback chain. Every model of one credential is tried before the next credential, and every credential of one provider before the next provider.

The ordering is not arbitrary. A rate-limit response from a shared free model queue often clears if a sibling model on the same credential is tried, while a daily account quota only clears with a different credential; trying every model of one credential before moving on serves both cases without confusing them.

3.3.4 The Conversational Gap-Closing Loop

The loop is the part of the system that exists because transcription is unreliable. Its shape is shown in Figure 5.

Extraction fills ten fixed fields: the main problem, when it started and for how long, the details of the symptom, associated symptoms, relevant medical history, current medicines, allergies, recent changes or exposures, treatments already tried, and the patient's own current concern. Each field carries its text in both English and Bangla, filled in the same extraction call so that bilingual display

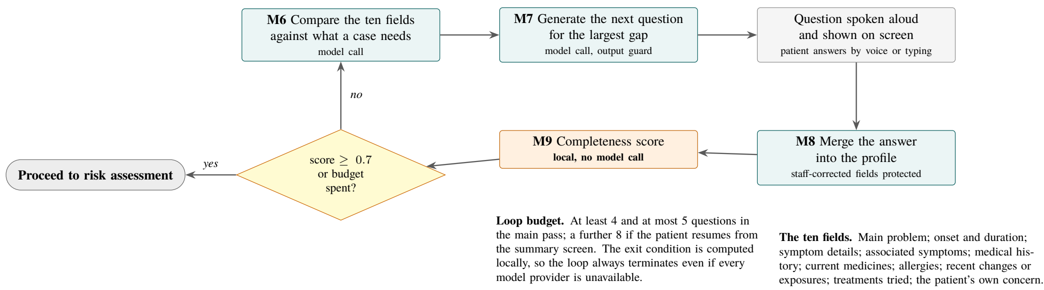


Figure 5. The gap-closing loop. Completeness is computed locally over ten fixed fields, and the loop exits on the completeness threshold or the question budget, whichever comes first.

costs no extra model quota, and each carries a provenance marker recording whether the value came from the system or from a member of staff.

Module 9 computes completeness locally as the fraction of those ten fields that hold text in any language slot. No model is involved, so the loop's exit condition cannot fail because a provider is down. The loop asks at least four questions and at most five, and exits when completeness reaches 0.7 or when the budget is spent. The floor of four exists because a patient who gives one rich opening sentence can cross the threshold immediately, and stopping there would waste the opportunity to ask about medicines and allergies. A separate budget of eight questions covers the case where a patient returns to the summary screen and resumes.

Module 8 merges each answer back into the profile. Any field a member of staff has already corrected is protected and is never overwritten by a later model pass, because a human reading the patient's own words is a better authority than a model reading a noisy transcript.

Module 7 has an output guard. A generated question that strays into dosing or prescribing is rejected, and a deterministic replacement question is asked instead, so a rejected question never costs the patient their turn.

3.3.5 Risk Assessment and the Safety Layer

Module 10 combines two independent judgements, and the order matters.

First, a local rule scans every patient utterance, both the raw text and the corrected text, for danger-sign phrases. The rule is a plain substring match over a fixed list, written out in Bangla, in romanised Bangla and in English, so that dialect particles around a phrase do not defeat it. The five categories and examples of their trigger phrases are given in Table IV. The rule is tuned for recall rather than precision, on the reasoning that a false high tier costs a doctor's attention while a missed one could cost a life.

TABLE IV. Risk tiers and the rule-based danger-sign categories.

Category	Examples of trigger phrases (Bangla, romanised, English)
Chest pain	বুকে ব্যথা, বুকে চাপ; *buke betha*, *buke chap*; "chest pain", "chest pressure"
Severe breathing difficulty	শ্বাসকষ্ট, দম বন্ধ; *shash koshto*, *dom bondho*; "cannot breathe", "shortness of breath"
Stroke signs	মুখ বেঁকে, কথা জড়িয়ে, এক পাশ অবশ; *mukh beke*, *kotha joriye*; "face drooping", "slurred speech"
Loss of consciousness	অজ্ঞান, জ্ঞান হারিয়ে; *oggan*, *gyan hariye*; "unconscious", "fainted"
Severe uncontrolled bleeding	প্রচুর রক্ত, রক্ত বমি, কাশিতে রক্ত; *onek rokto*; "bleeding won't stop", "coughing up blood"

Tiers: low, medium, high, critical. A triggered category forces *critical* regardless of the model's answer. A failed model call defaults to *medium*, never to *low*.

Second, a language model classifies the case into one of four tiers and lists the observable factors behind its choice. It is instructed not to name diseases.

The two are then combined. If any danger sign was found, the tier is forced to critical. If the model call failed or could not be parsed, the tier defaults to medium rather than low, because Rule 3 forbids a failure from producing a reassuring answer. The rule can only ever raise concern; it can never lower it, and a member of staff cannot downgrade a rule-forced critical tier.

Module 11 writes a plain-language reason for every assessment. If that call fails, a deterministic reason is composed from the triggered categories and the recorded factors, so no risk row is ever stored without an explanation attached.

3.3.6 Identity, Audit and Privacy

Patients identify themselves with a phone number and a one-time code. The code is six random digits. Only a salted hash of it is stored, it expires after five minutes, it can be used once, it is compared in constant time, the account locks after five failed attempts, and resending is throttled. The sending channel sits behind an interface with two implementations: a development channel that writes the code to the server log, and a real short-message channel through an open-source gateway running on the team's own subscriber module. A universal development bypass code exists, but it is honoured only inside the development channel and only when an explicit flag is set, so it is structurally impossible to reach under a production channel.

Every state change is written to an append-only audit table, including values written by the system itself. This closed a real problem found during testing: a patient's name filled in automatically by the extraction stage used to enter a permanent medical record with no trace of where it came from. The name now carries its origin, derived from the audit trail with no new column, and an origin that cannot be established is reported as unknown rather than guessed.

Two limits should be stated plainly here rather than left to the conclusions. Staff authentication is stubbed: a member of staff selects a seeded account instead of logging in, and the user table and roles exist as the seam for real authentication later. There is no encryption at rest and no transport encryption configured. Both are necessary before the system could touch real patient data, and neither was built.

3.3.7 The Three Portals

All three portals share one stylesheet with a common set of design tokens, one script holding the single mapping from internal tier codes to display labels, and one text-to-speech helper. Bilingual English and Bangla display is done with paired data attributes on each element and a single language-switching function, so a translation exists in exactly one place. Bangla text is always set in a Bengali-capable font, never in a monospaced font, because monospaced fonts break Bengali shaping.

The two staff portals additionally load a depth and motion layer, which the kiosk never loads. Every animation in it sits behind a reduced-motion preference query, nothing is conveyed by movement alone, and a looping animation is reserved for urgency. One rule in that layer is worth recording because it came from a real defect: a card containing interactive controls is never moved. A card that moved in three dimensions on hover was rescaled by the perspective transform, so a button slid out from under a stationary pointer between the press and the release and the click landed on the wrong element. Depth is now carried by shadow alone.

Figures 6 and 7 show the patient kiosk during the conversation and at the review screen, Figure 8 shows the medic portal, and Figure 9 shows the doctor portal.

3.3.8 Clinical Documentation and Record Export

The database is the source of truth and every export is regenerated from it on request, so a re-export always reflects the latest staff corrections. Four export kinds exist: the verbatim transcript and the summary report as editable documents, the encounter as a standard interchange record, and the same record as a printable file.

Figure 10 shows one such record. The interchange export builds the whole encounter as one document bundle in the HL7 FHIR R4 format [27], using eleven resource types and ten sections, each section titled in both English and Bangla. It is claimed here as structurally valid and

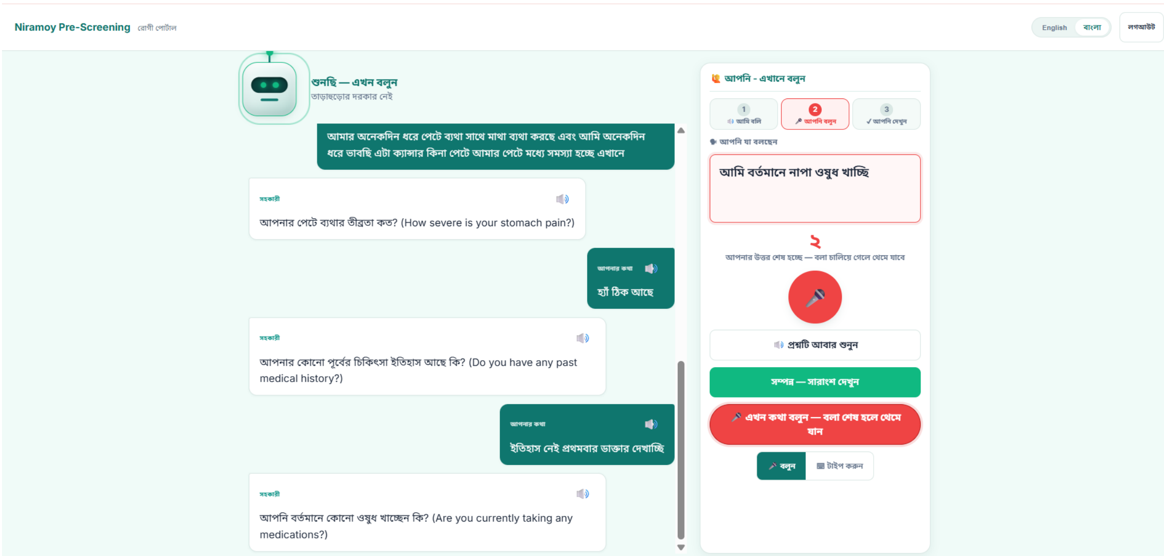


Figure 6. The patient kiosk during the follow-up conversation. Each question is shown in Bangla with an English gloss and is spoken aloud; the countdown on the right offers to close the turn and is cancelled if the patient speaks again. The Speak and Type controls sit together at the bottom.

conservative in what it asserts; it has not been certified, it does not claim conformance to any implementation guide, and it has never been sent to a real receiving system. The system's own suggested condition is excluded from it entirely, because the disclaimer that accompanies that suggestion on screen would not survive being ingested somewhere else. The doctor's own diagnosis, typed by the doctor, is exported.

The printable export renders that same bundle and never reads the database again, so the two exports cannot drift apart. It required a text shaping engine because the usual Python document library cannot render Bengali conjuncts correctly, and mangled Bangla in a medical record would be a Rule 1 failure. The renderer refuses to produce a file rather than shipping bad Bangla, and a test checks every character it draws against the font, because a missing glyph disappears silently instead of raising an error.

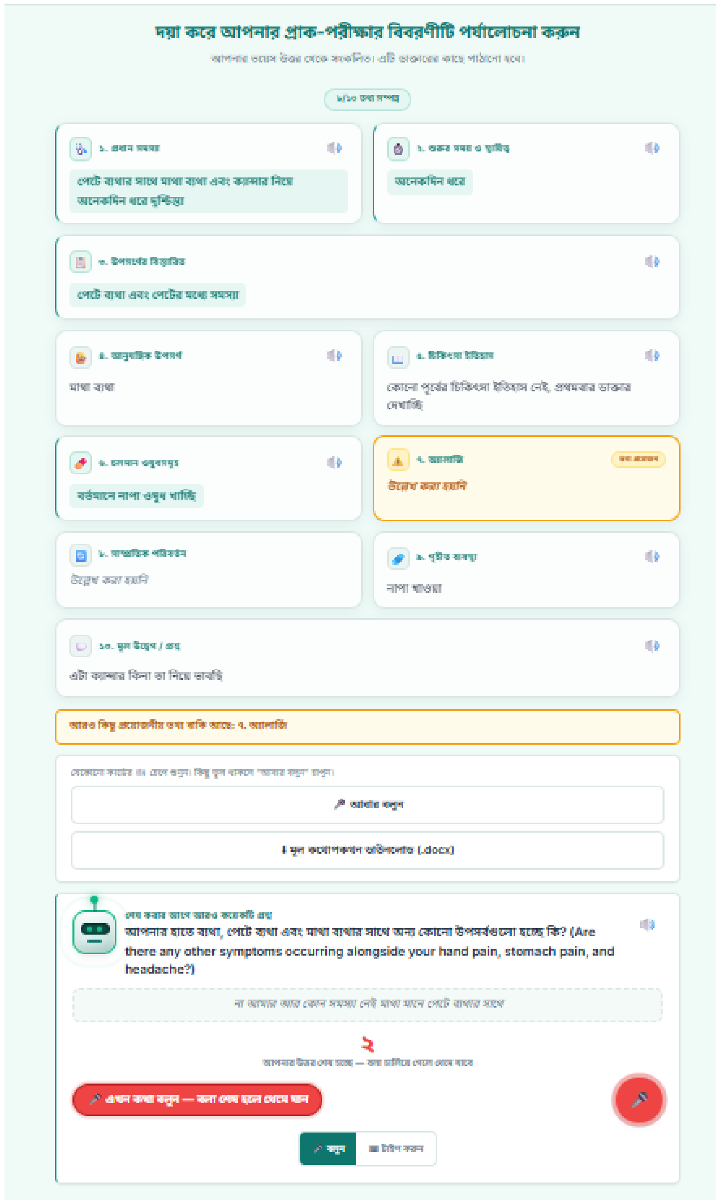


Figure 7. The summary the patient reviews before submitting, showing the extracted fields in their chosen language.

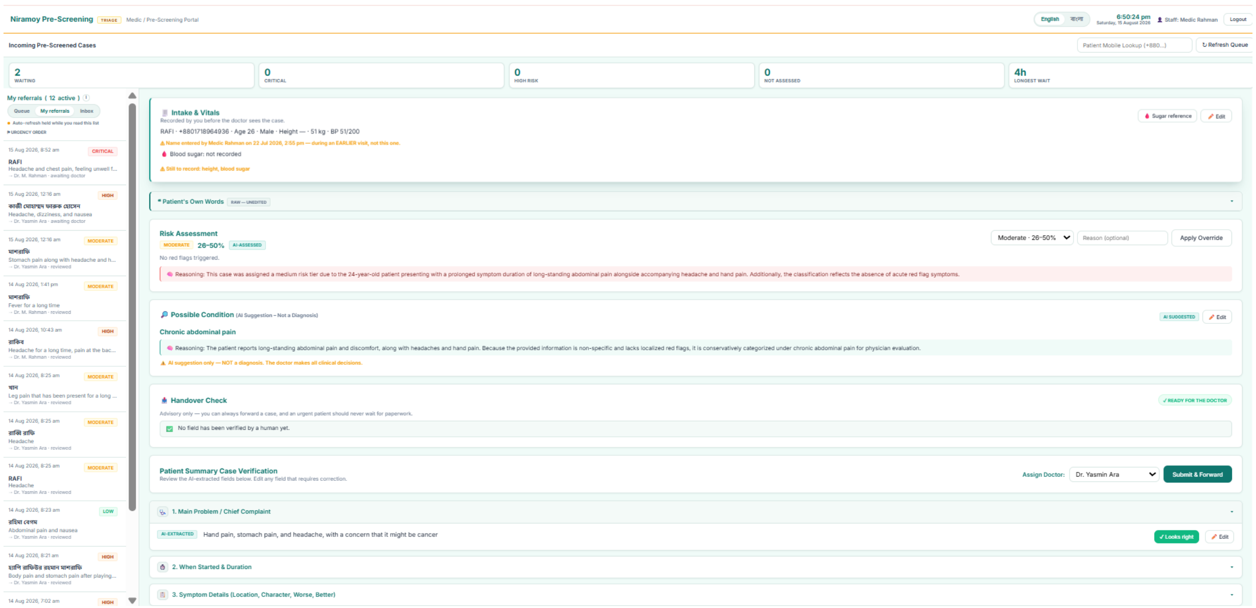


Figure 8. The medic portal. The queue on the left is ordered by urgency and not by arrival time. The case shows intake and vital signs, the risk assessment with its reason, the suggested condition with its disclaimer, the advisory handover check, and the ten fields for verification against the patient's own words.

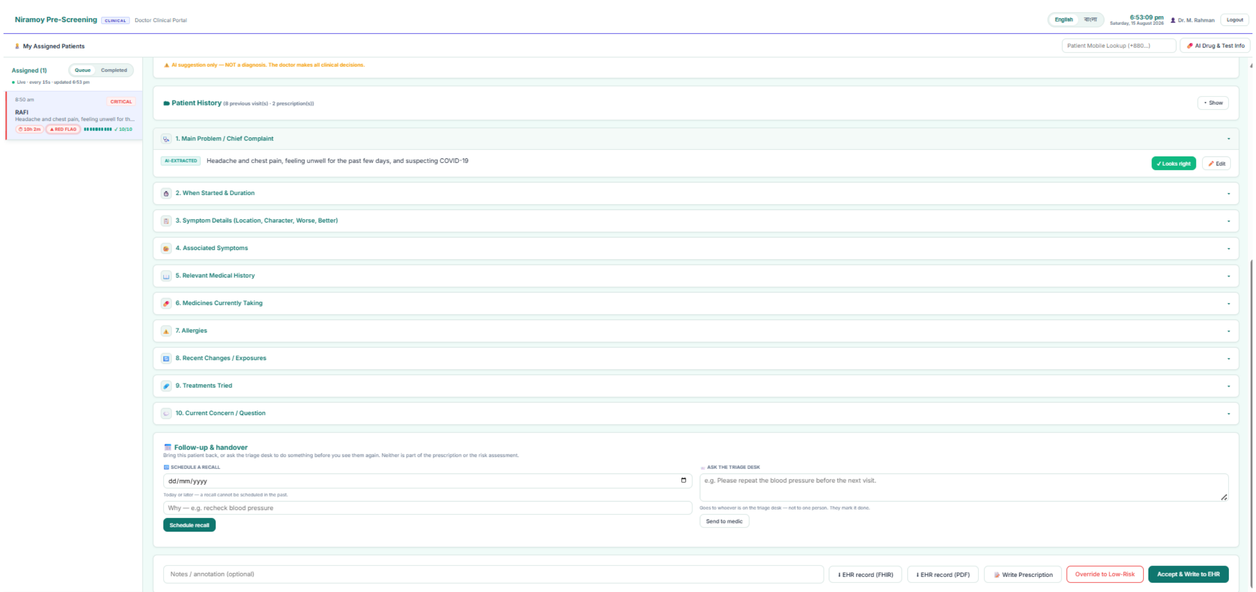
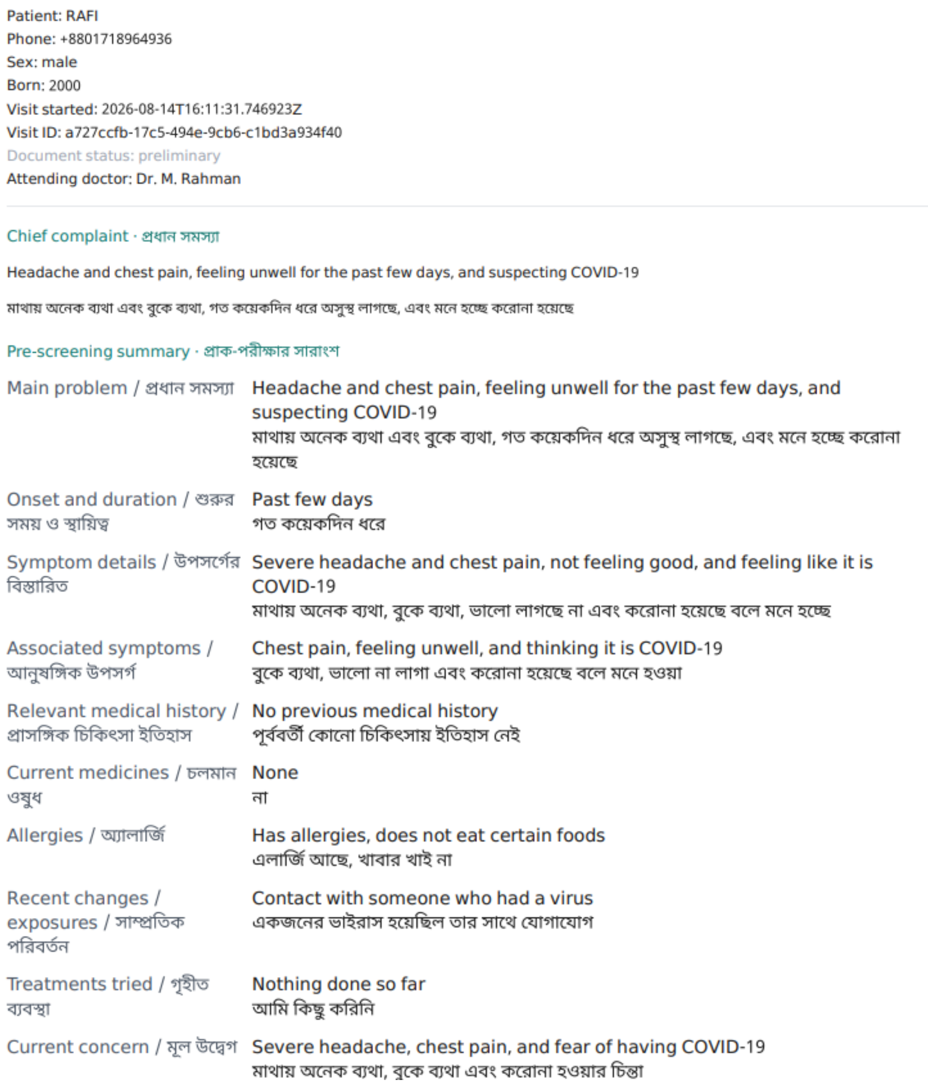


Figure 9. The doctor portal. The safety information appears first, followed by the patient's history, the ten fields, the follow-up and handover controls, and the record export and prescription actions along the bottom.



Patient: RAFI
Phone: +8801718964936
Sex: male
Born: 2000
Visit started: 2026-08-14T16:11:31.746923Z
Visit ID: a727ccfb-17c5-494e-9cb6-c1bd3a934f40
Document status: preliminary
Attending doctor: Dr. M. Rahman

Chief complaint · প্রধান সমস্যা

Headache and chest pain, feeling unwell for the past few days, and suspecting COVID-19

মাথায় অনেক ব্যথা এবং বুকে ব্যথা, গত কয়েকদিন ধরে অসুস্থ লাগছে, এবং মনে হচ্ছে করোনা হয়েছে

Pre-screening summary · প্রাক-পরীক্ষার সারাংশ

Main problem / প্রধান সমস্যা	Headache and chest pain, feeling unwell for the past few days, and suspecting COVID-19 মাথায় অনেক ব্যথা এবং বুকে ব্যথা, গত কয়েকদিন ধরে অসুস্থ লাগছে, এবং মনে হচ্ছে করোনা হয়েছে
Onset and duration / শুরুর সময় ও স্থায়িত্ব	Past few days গত কয়েকদিন ধরে
Symptom details / উপসর্গের বিস্তারিত	Severe headache and chest pain, not feeling good, and feeling like it is COVID-19 মাথায় অনেক ব্যথা, বুকে ব্যথা, ভালো লাগছে না এবং করোনা হয়েছে বলে মনে হচ্ছে
Associated symptoms / আনুষঙ্গিক উপসর্গ	Chest pain, feeling unwell, and thinking it is COVID-19 বুকে ব্যথা, ভালো না লাগা এবং করোনা হয়েছে বলে মনে হওয়া
Relevant medical history / প্রাসঙ্গিক চিকিৎসা ইতিহাস	No previous medical history পূর্ববর্তী কোনো চিকিৎসায় ইতিহাস নেই
Current medicines / চলমান ওষুধ	None না
Allergies / অ্যালার্জি	Has allergies, does not eat certain foods এলার্জি আছে, খাবার খাই না
Recent changes / exposures / সাম্প্রতিক পরিবর্তন	Contact with someone who had a virus একজনের ভাইরাস হয়েছিল তার সাথে যোগাযোগ
Treatments tried / গৃহীত ব্যবস্থা	Nothing done so far আমি কিছু করিনি
Current concern / মূল উদ্বেগ	Severe headache, chest pain, and fear of having COVID-19 মাথায় অনেক ব্যথা, বুকে ব্যথা এবং করোনা হওয়ার চিন্তা

Figure 10. The interchange record generated for one visit, shown in the doctor portal.

Chapter 4 Investigation, Result, Analysis and Discussion

This chapter reports four different kinds of evidence, and it keeps them apart on purpose. Sections 4.1 to 4.4 report a controlled benchmark of external speech models, which produced the project's only quantitative results. Section 4.5 reports the verification carried out on the system that was built, which establishes what the software does but not how accurate it is in clinical use. Section 4.6 states, explicitly, what was not measured at all. Merging these into a single accuracy claim would misrepresent all of them.

4.1 The Multi-Dialect Bangla Speech Corpus

No public dataset covers Bangladeshi regional dialects speaking informally about health, so a corpus was assembled for the benchmark. Audio in five regional varieties was collected from publicly available recordings and organised under a fixed naming convention recording the dialect, an index, the speaker's gender and an approximate age band. Each file's metadata, including its duration measured directly from the audio, was logged in a dataset register. Table V gives the composition.

TABLE V. Composition of the multi-dialect Bangla speech corpus.

Variety	Code	Recordings	Duration
Puran Dhaka (Dhakaiya)	pd	1	35.7 min
Barishal	br	2	33.8 min
Sylheti	sy	2	31.7 min
Standard Bangla	nb	3	87.0 min
Indian Bangla	ib	2	95.9 min
Total		**10**	**284.1 min (≈ 4.7 h)**

Ten speakers appear across the recordings, eight male and two female, with ages spanning the twenties to the fifties. The recordings include both monologue and dialogue, and they contain natural code-mixing, with English words for medicines, body parts and time expressions embedded in Bangla sentences. All audio comes from publicly available sources; no clinical patient recording was used at any point.

Preprocessing converted every file to 16 kHz mono, normalised the loudness, and split the recordings into ten to fifteen second clips at speech boundaries found by a voice-activity detector [23].

Segmentation is necessary because most of the evaluated models cannot accept long-form input directly. This yields roughly 1,100 segments in total, from which the first fifty clips per variety were taken as the evaluation subset, giving 250 clips.

Reference transcripts for those clips were written by hand. This is where the first limitation of the evaluation appears, and it is stated here rather than buried later. Of the 250 clips in the evaluation subset, only **147** could be paired with a reference transcript at scoring time. The 147 cover Barishal (47 clips), Standard Bangla (50) and Indian Bangla (50). **Puran Dhaka and Sylheti contributed no scored clips at all.** Reference files for those two varieties do exist — a later check of the reference folder found 44 Puran Dhaka and 50 Sylheti transcript files — but their filenames did not line up with the names of the processed audio clips, so the scoring step could not pair them. Every error rate reported in this chapter is therefore computed over 147 clips from three varieties, and nothing at all is reported for the two varieties that are furthest from standard Bangla. Fixing the filename matching and rescoring is listed in Section 8.3 as future work.

4.2 Experiment 1: Baseline Speech Recognition Benchmark

4.2.1 Method

Twelve open-source Bangla and multilingual speech recognition models were evaluated. They span the main families in current use: four wav2vec 2.0 and XLSR variants fine-tuned on Bengali, four Whisper sizes from OpenAI, two Bengali fine-tuned Whisper checkpoints, Meta's massively multilingual model, and Meta's SeamlessM4T. All were loaded through one common model interface [22] and run on free-tier cloud graphics processors, since the team's own machines have none.

Each model transcribed all 250 clips. The resulting hypotheses were scored against the reference transcripts wherever a reference could be paired, using three metrics. Word Error Rate and Character Error Rate were computed with the `jiwer` library [25] after light normalisation of whitespace, punctuation and Unicode form. BLEU [24] was computed with the evaluation library from the same model ecosystem. Word Error Rate is the number of substituted, deleted and inserted words divided by the number of words in the reference; lower is better for the two error rates, higher is better for BLEU.

One property of Word Error Rate matters for reading the tables that follow. The denominator is the length of the reference, but insertions in the numerator have no upper bound. A model that hallucinates and produces far more words than were actually spoken can therefore score above 1.0.

This is not a scoring artefact to be explained away; it is itself a finding, because it identifies the models that fail by inventing text rather than by staying silent.

4.2.2 Results

Table VI gives the results for all twelve models, sorted by Word Error Rate.

TABLE VI. Baseline speech recognition results on the multi-dialect Bangla corpus.

Model	WER ↓	CER ↓	BLEU ↑
BengaliAI Regional	**0.4694**	**0.2429**	**0.2730**
BengaliAI Whisper	0.5055	0.3075	0.2527
MMS [4]	0.5568	0.2687	0.2003
Vakyansh-BN [14]	0.6142	0.2950	0.1810
Wav2Vec2 XLSR-CV-BN [5]	0.6808	0.4365	0.1668
Wav2Vec2 XLSR-300M-BN [6]	0.7023	0.3894	0.1191
SeamlessM4T [7]	0.7679	0.5210	0.1233
Whisper Large-v3 [3]	0.9097	0.5827	0.0646
Wav2Vec2 XLSR-53-BN	1.0023	0.8245	0.0002
Whisper v3 Turbo	1.0264	0.7850	0.0117
Whisper Medium	1.1680	1.2299	0.0002
Whisper Small	1.9580	1.7060	0.0000

Computed over the 147 clips for which a reference transcript could be paired, covering Barishal, Standard Bangla and Indian Bangla. Best value in each column in bold.

The best model, a Bengali regional fine-tune of Whisper Medium, reached a Word Error Rate of 0.4694. In plain terms, it got roughly 47 out of every 100 words wrong. Its Character Error Rate of 0.2429 is much lower, which tells us the output is usually recognisable Bangla with the wrong words rather than nonsense. Its BLEU of 0.2730 is low in absolute terms but is the highest in the set.

At the other end, the general Whisper sizes without Bengali fine-tuning are unusable on this material. Whisper Small scores 1.9580, meaning it produced roughly twice as many errors as there were words to get right, which only happens when a model repeats or invents long stretches of text. That behaviour is dangerous in a medical context in a specific way: a model that stays silent produces an obvious gap, while a model that invents fluent text produces a plausible-looking record that is wrong.

4.3 Experiment 2: Larger Multimodal Audio Models

4.3.1 Method

Six larger models, from roughly 1.7 to 7 billion parameters, were evaluated on the same clips with the same metrics, to test whether scale alone closes the dialect gap. Qwen2-Audio [8] was run in two configurations: transcribing directly to Bangla, and transcribing through English. Qwen3-ASR, Voxtral-Mini [9], Phi-4 Multimodal [10] and Qwen2.5-Omni completed the set.

4.3.2 Results

Table VII gives the results.

TABLE VII. Larger multimodal audio model results, against the best baseline.

Model	**WER ↓**	**CER ↓**	**BLEU ↑**
BengaliAI Regional (best baseline)	*0.4694*	*0.2429*	*0.2730*
Qwen2-Audio (via English)	1.0039	0.9214	0.0001
Qwen2-Audio (direct)	1.0837	0.9327	0.0003
Qwen3-ASR	1.1339	0.8962	0.0031
Voxtral-Mini	1.3627	0.9722	0.0626
Phi-4 Multimodal	*no usable output produced*		
Qwen2.5-Omni	*no usable output produced*		

Two results need stating carefully.

First, every model that produced output scored a Word Error Rate above 1.0. All four were worse than every Bengali-specialised baseline in Table VI, and the closest of them trails the best baseline by more than 53 percentage points. Inspecting the stored hypotheses shows why: the audio language models frequently answered in English, or replied with a sentence about the request instead of a transcription. One stored hypothesis for a Puran Dhaka clip reads "Sorry, but I don't see any audio attached to your message." These are instruction-following failures rather than acoustic ones, but from the point of view of a pre-screening system the outcome is the same.

Second, **Phi-4 Multimodal and Qwen2.5-Omni produced no usable output at all**, and no score is reported for them. Their rows are empty in every stored result file. It would have been possible to record them as a Word Error Rate of 1.0, or of 0, and either would have been an invention. They are reported here as failures because that is what the evidence shows.

Figure 11 places every scored model from both experiments on one axis.

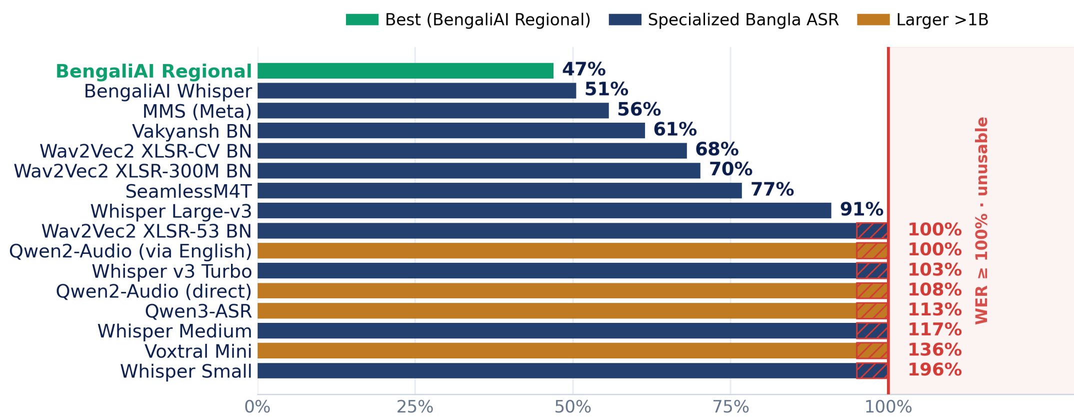


Figure 11. Word Error Rate of every scored model from both experiments. Models at or beyond 1.0 produced more errors than there were reference words.

4.4 Analysis by Dialect

Figure 12 breaks the six strongest baseline models down by variety, and Table VIII gives the same numbers.

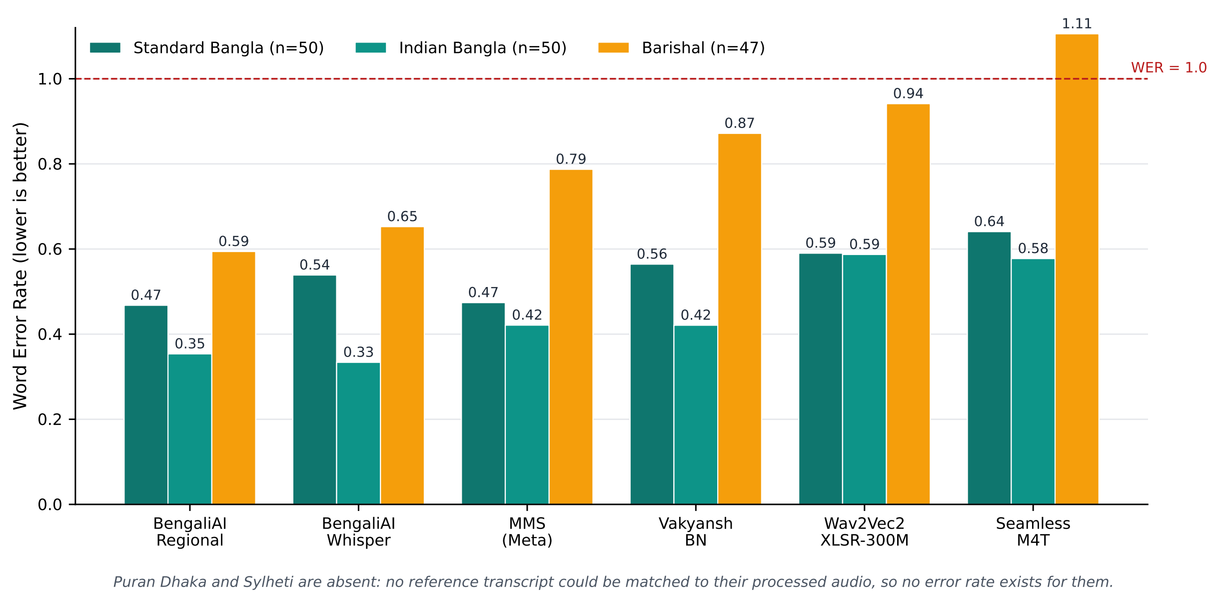


Figure 12. Word Error Rate by variety for the six strongest baseline models.

The pattern is consistent. Every model does best on Indian Bangla, which is closest to the broadcast-standard speech these models were trained on, slightly worse on Standard Bangla, and clearly worst on Barishal. For the best model the gap between Standard Bangla and Barishal is about 13 percentage points; for SeamlessM4T it is about 46 points, and the Barishal score passes 1.0. The weaker the model's Bengali specialisation, the more the dialect costs it.

TABLE VIII. Word Error Rate by variety for the leading baseline models.

Model	Standard (n = 50)	Indian (n = 50)	Barishal (n = 47)
BengaliAI Regional	0.4678	0.3537	0.5942
BengaliAI Whisper	0.5389	0.3338	0.6526
MMS	0.4741	0.4211	0.7873
Vakyansh-BN	0.5644	0.4209	0.8717
Wav2Vec2 XLSR-300M-BN	0.5902	0.5872	0.9415
SeamlessM4T	0.6409	0.5776	1.1057

No values are reported for Puran Dhaka or Sylheti. Reference transcripts for those two varieties could not be paired with the processed audio, so no error rate was computed for them.

A second view of the same data is worth reporting. Counting a clip as correctly recognised only when its Word Error Rate falls below 0.3, the best model recovers 19 of 50 Indian Bangla clips and 3 of 50 Standard Bangla clips, and **0 of 47 Barishal clips**. No model in the benchmark cleared that threshold on a single Barishal clip. That is a stark result and it is exactly the kind of result that matters clinically, because a clip with 60 per cent of its words wrong is not a transcript a doctor can safely read.

It must be repeated that this analysis covers one regional dialect only. Whether Puran Dhaka and Sylheti behave like Barishal, better or worse, is not known from this work.

4.5 Verification of the Implemented System

The evidence in this section is of a different kind from the benchmark above. It establishes that the implemented software behaves as specified. It does not establish clinical accuracy, and it is not presented as if it did.

4.5.1 The Automated Test Suite

The system is covered by an automated test suite of 89 test files. The last recorded full run of the suite, on 19 August 2026, reported **1,196 tests passed, 2 skipped and 0 failures**. All model calls in the suite are replaced with fakes and every database is either in memory or a throwaway file, so the suite spends no external quota and touches no real data.

The suite was written around the four rules rather than around code coverage, so that a future change which breaks a rule fails the build. Table IX summarises what each area of testing establishes and, just as importantly, what it does not.

TABLE IX. Verification activities and what each one does and does not establish.

Area	What is checked	What it does not show
Rule enforcement	Raw text is never updated; an answer submission returns the value unchanged; the follow-up loop never repeats an answered item and always terminates.	That the transcript itself is correct.
Safety	Every phrase in the danger-sign list forces the critical tier; a failed model call defaults to medium; staff cannot downgrade a rule-forced critical.	That the phrase list is clinically complete.
Pipeline	Extraction fills the ten fields; bilingual values are produced; completeness scoring and the question budget behave as specified.	Extraction precision or recall on real speech.
Provider resilience	The multi-credential, multi-model chain is walked in the specified order; a rate limit cools down only that combination; no credential is ever written to a log.	Behaviour against real providers under real quota pressure; all provider tests use fakes.
Migrations	Each schema change is tested on a fresh database, as an in-place upgrade, and as a downgrade, including database-level constraints.	—
Record export	The interchange bundle is structurally coherent and every internal reference resolves; the printable file reproduces the bundle; every drawn character exists in the font.	Interoperability with any real receiving system, which was never attempted.
Portals	Queue ordering, doctor history, kiosk voice flow, one-time-code entry and speech fallback behave as specified.	How the portals look. Appearance was checked by hand, not by automated comparison.

4.5.2 The Supervised Live Session

Automated tests cannot exercise a microphone. One supervised live session was therefore carried out on 12 July 2026 on a Windows 11 machine in Google Chrome, against a real server instance and using synthetic patient details only. Five prepared test cases were run: raw transcript capture, spoken and displayed follow-up questions, answering entirely by voice, the follow-up loop, and a danger-sign phrase forcing the critical tier. All five passed. The operator recorded that recognition was accurate, that the delay from speech to text was around two seconds, that the questions were spoken correctly, and that the follow-up questions were relevant.

That paragraph is the complete evidence from the session, and it is qualitative. No Word Error Rate was computed by hand, no labelled test set was used, no timing was instrumented, and the session was run on one machine by one operator. It confirms that the voice loop works end to end. It is not a measurement of how well it works, and this report does not use it as one.

4.6 What Was Not Measured

The following were not measured at any point during the project. They are listed in full because a reader is entitled to know the boundary of the evidence.

- **Word Error Rate of the deployed recogniser.** The benchmark measured candidate open-source models. The system in the demonstration uses the browser's speech recognition service, which was never formally benchmarked. The 0.4694 figure does not describe it.
- **Extraction precision and recall.** How often the ten clinical fields are filled correctly from a real transcript is unknown. The tests check that the fields are filled, not that they are right.
- **Risk tier accuracy.** There is no labelled set of cases with an agreed correct tier, so the classifier's agreement with clinical judgement is unmeasured. Only the deterministic rule's behaviour is verified.
- **Latency.** Beyond the operator's impression of about two seconds in one session, no timing distribution was recorded for speech capture, model calls or the end-to-end visit.
- **Provider failover under real conditions.** The chain has only been exercised against fakes.
- **Usability.** No user study, no task-completion measurement, and no testing with elderly or low-literacy participants, who are the intended users. Ethical clearance would be required for this and was not sought.
- **Clinical benefit.** No claim is made about consultation time saved, documentation quality or patient outcome. Establishing any of these would require a supervised deployment.

4.7 Discussion

The benchmark produced two findings, and the second was the more surprising one.

The first is that Bangla dialect speech recognition is genuinely unsolved by the openly available models. Even the best Bengali-specialised model loses about half the words, and on the one dialect for which a measurement exists it recovers no clip cleanly at all. This is not a small gap to be closed by tuning.

The second is that scale did not help. Every large multimodal audio model tested was worse than the small specialised ones, two failed outright, and several answered in the wrong language rather than transcribing. The intuition that a bigger model trained on more of the internet will be better at a low-resource language did not hold here. A plausible reason is that these models are trained and instruction-tuned overwhelmingly on high-resource languages, so their instruction-following behaviour dominates their acoustic behaviour when the input is unfamiliar. Whatever the cause, the practical conclusion for a project like this one is that waiting for a larger model is not a strategy.

Together these two findings determined the architecture described in Chapter 3. If the transcript cannot be trusted, then four things follow directly. The patient's exact words must be preserved permanently, so that a human can always check what was actually captured. Cleaning and structuring must happen in separate, separately stored steps, so that an error introduced downstream never destroys the original. The information that recognition loses must be recovered by asking, which is what the follow-up loop does. And the safety check must not depend on any model at all, which is why the danger-sign rule is a local substring match that runs whether the language models answer or not, and why a failed classification defaults to medium rather than to low.

The honest summary of the project's evidence is this. What was measured is that existing open speech models are not accurate enough on Bangladeshi dialect speech, and that larger models do not fix it. What was built is a system designed around that measurement, whose behaviour is protected by a large automated suite and which has been demonstrated end to end once with a real microphone. Whether the built system improves clinical documentation in practice is not something this project has measured, and Section 8.3 sets out what would have to be done to find out.

Chapter 5 Impacts of the Project

This chapter separates two things that are easy to blur. The system was *designed* to produce certain benefits, and the design reasoning for each is given below. None of these benefits has been *measured*, because that would require a supervised deployment with real patients and the ethical clearance that goes with it. Where a statement is about intended effect rather than observed effect, it is written that way.

5.1 Impact on Societal, Health, Safety, Legal and Cultural Issues

5.1.1 Societal and Health

The clearest intended benefit is access. A patient who cannot read or write, or who is elderly and uncomfortable with a touchscreen keyboard, is excluded by every intake form that expects typing. Speaking is the one interface almost everybody already has. The system is built voice-first for that reason: the assistant speaks its question aloud, the microphone opens by itself, and the patient simply answers. Typing remains available beside the microphone, because a patient should never be trapped by a failed microphone or a noisy room, but it is the alternative and not the default.

The second intended benefit is language. Existing digital health tools in Bangladesh are largely English-first, which asks the patient to describe a medical problem in a language they may not use at home. This system takes Bangla, Banglish and regional dialect speech as normal input rather than as an edge case, and presents everything back in both Bangla and English so that the patient and the clinician can each read the version they prefer.

The third is time. Symptom collection happens while the patient is waiting rather than during the consultation. If this works as designed, the doctor begins already informed, with the history, medicines and allergies already on screen, and can spend the short consultation on examination and decision instead of on transcription. This is the benefit most often claimed for systems of this kind and it is the one this project has *not* measured; establishing it would need timing data from a real clinic.

There is also a documentation benefit that does not depend on any of the above. Most small clinics in Bangladesh keep no structured records at all. Every encounter this system handles produces a stored, searchable record with the patient's own words attached, which means a returning patient's earlier visits and prescriptions can be seen rather than re-asked.

5.1.2 Safety

A pre-screening tool sits close enough to clinical decision-making to be dangerous if it is careless, and the safety posture of the system is deliberate.

The system never diagnoses. Where it does surface a possible condition for staff, it is labelled as a suggestion, it carries a disclaimer attached on the server so that no client can strip it, it never pre-fills the doctor's diagnosis field, and it is excluded from the exported clinical record entirely, because a disclaimer shown on screen would not survive being ingested by another system.

The system never reassures falsely. The danger-sign rule is deterministic, runs locally, and forces the highest tier regardless of what the language model concluded. A failed model call produces a medium tier, never a low one. A member of staff can raise a tier but cannot lower one that the rule forced.

The system never hides what it did. Every risk assessment carries a stored plain-language reason, every state change is written to an append-only audit trail, and a value written automatically by the system is audited in the same way as one typed by a person, so a name that appeared without a human typing it can be traced rather than assumed.

Finally, the system is honest about the transcript. The patient's exact words are kept unchanged and shown to staff read-only beside the extracted fields, precisely so that a member of staff can catch a misrecognition instead of trusting a clean looking summary. Given a Word Error Rate near 0.5, this is the single most important safety property in the design.

5.1.3 Legal

Bangladesh does not yet have a comprehensive data-protection statute in force comparable to the frameworks of the European Union or of some neighbouring countries, and the regulatory position for clinical data held by a small private clinic is correspondingly unsettled. This project has therefore taken the conservative route rather than waiting for a rule to comply with.

Only synthetic or consented sample data was used at any point in development. No real patient recording, name or clinical detail entered the system. This was not a formality: the speech recogniser sends audio to a cloud service, and the free model tiers used for the language tasks may train on what is sent to them, so processing real patient speech through the system as it stands would be indefensible.

Three legal prerequisites for real use are missing and are stated as such. Staff authentication is stubbed rather than implemented, so there is no enforceable access control. There is no encryption of data at rest or in transit. There is no consent-management interface or retention policy, although a consent flag exists in the schema as the place one would attach. Any real deployment would need all three, together with a provider that contractually does not train on submitted data.

5.1.4 Cultural

Two cultural considerations shaped the interface. First, the system treats dialect as legitimate speech rather than as an error to be corrected. Converting a patient's spoken Banglish directly into formal Bangla script can quietly change what they said, so the system stores the utterance as it was recognised and does its normalisation later and separately. Second, everything the patient and the staff see is bilingual, with Bangla rendered in a proper Bengali typeface rather than a monospaced font that would break the conjuncts. These are small choices, but a medical interface that displays a patient's own language badly does not read as trustworthy.

5.2 Impact on Environment and Sustainability

5.2.1 Environmental Footprint

The computational footprint of this project is small, and that is a design outcome rather than an accident.

No model was trained. Both benchmark experiments were inference-only, run on free-tier cloud graphics processors for a bounded number of hours. Training a speech model from scratch, or fine-tuning a large one fully, would have consumed orders of magnitude more energy; the planned future adaptation deliberately uses a parameter-efficient method [21] that trains a small number of adapter weights instead of the whole model.

The delivered system itself runs on ordinary hardware with no dedicated graphics processor. The backend is a single process with a single-file database, and the three portals are static files with no build step and no framework bundle to download. A clinic can run the whole thing on the kind of desktop it already owns, which avoids both the manufacture of new hardware and the continuous draw of a server.

The one genuine environmental cost is the external model calls, since each one runs on somebody else's data-centre hardware. The system limits this structurally rather than by good intentions: the completeness check that decides whether to ask another question runs locally with no model call, the danger-sign rule is local, the report is assembled locally, and the question budget caps the number of model calls per visit.

5.2.2 Sustainability of the Approach

Sustainability here also means whether the approach could keep running in a Bangladeshi clinic after a capstone project ends.

The system is built entirely from free and open-source components, so there is no licence to renew. Every external service sits behind a swappable interface, so a provider that withdraws its free tier can be replaced by configuration rather than by rewriting code, and the same seam is the path to a fully local model later. The database can be moved from a single file to a full database server by changing one connection string. Nothing in the design assumes a permanent relationship with any particular vendor.

Two sustainability weaknesses should be named alongside this. The system currently depends on internet connectivity for both speech recognition and the language tasks, which is a real constraint in rural clinics; moving recognition to a local model is the first item of future work for exactly this reason. And free tiers are not a durable foundation for a production service, which is why the provider chain treats quota exhaustion as an expected condition rather than an error.

5.2.3 Sustainable Development Goals

The project relates most directly to two of the United Nations Sustainable Development Goals. It addresses **Goal 3, Good Health and Well-being**, by aiming to improve the quality and continuity of clinical documentation in settings that currently have none. It addresses **Goal 10, Reduced Inequalities**, by building for patients who are excluded by literacy, by age or by dialect from the digital health tools that already exist. As with everything else in this chapter, the contribution is by design; it has not been measured.

Chapter 6 Project Planning and Budget

6.1 Project Planning

The project ran across two semesters with different characters. CSE499A was a measurement semester: collect a corpus, benchmark the available models, and find out how hard the underlying problem actually is. CSE499B was a build semester: turn that answer into a working system. Figure 13 shows both.

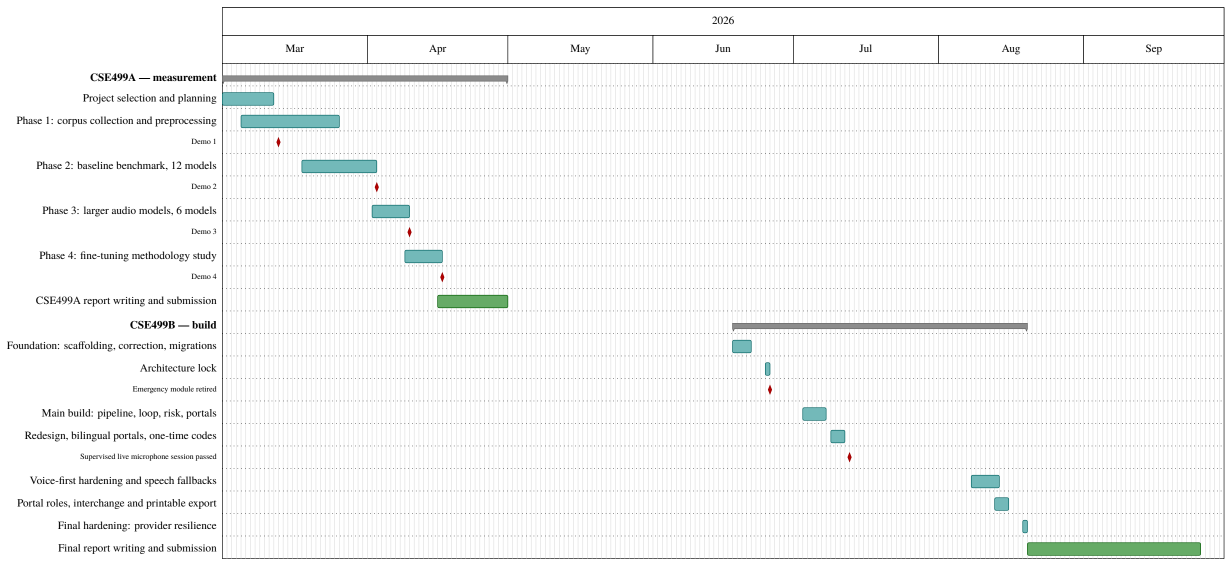


Figure 13. Execution timeline of CSE499A and CSE499B. The CSE499B phases are reconstructed from the dated entries of the project development log.

6.1.1 CSE499A — March to April 2026

The first semester was organised around five phases, each closed by a demonstration to the faculty advisor. Phase 1 collected and preprocessed the five-variety audio corpus. Phase 2 benchmarked twelve baseline speech recognition models. Phase 3 benchmarked six larger multimodal audio models on the same clips. Phase 4 studied the methodology for parameter-efficient fine-tuning of a mid-sized model; it remained a methodology study and no training was carried out. Phase 5 was selected near the end of the semester as a comparative study of general-purpose assistants on medical extraction, and its outcome fed directly into the redesign at the start of the second semester.

6.1.2 CSE499B — June to August 2026

The second semester was worked in short, dated sessions, each of which ended with an update to the project's own logs. Those logs are the source of the timeline below; the report does not claim dates that are not recorded in them.

18–21 June: foundation. Project scaffolding, the correction service, a browser-only speech path, document export, and versioned database migrations.

25 June: architecture lock. The design review that removed the standalone emergency detection module and moved its safety responsibility into the risk assessment stage as a local rule. This decision shaped everything after it.

3–7 July: the main build. The twenty-step build of the full system: the visit aggregate, the extraction and summary stages, the follow-up loop, risk assessment and explanation, the report assembly, and the first versions of the medic and doctor portals.

9–11 July: redesign and identity. A second cycle covering the visual redesign, bilingual portals, background assessment on submission, and the real one-time-code identity flow with hashed, expiring, single-use codes.

12 July: the supervised live session. The first end-to-end run with a real microphone. All five prepared test cases passed, which was the gate the module status board had been waiting on.

8–13 August: voice-first hardening. The hands-free voice loop, the speech-output fallback chain, the spoken answer read-back, the confirmation window, and the handling of recogniser error conditions.

13–15 August: roles and clinical output. The two staff portals were audited as roles rather than as screens, which produced the urgency-ordered triage queue, vitals capture before referral, the patient timeline and prescription history on the doctor side, the standard interchange export and the printable record.

19 August: final hardening. Four sessions in one day covering provider resilience, error-message containment, credential redundancy, and one interface defect in which a moving card made a button unclickable.

September 2026: report writing and submission.

6.1.3 Deviations from the Plan

Two deviations are worth recording. The fine-tuning phase planned in CSE499A was not executed in CSE499B; the build of the working system was prioritised, and fine-tuning remains future work.

Separately, one step of the voice loop covering silence re-prompting and permission recovery was specified but deliberately not built, and is recorded as unbuilt in the project's own logs rather than quietly dropped.

6.2 Project Budget

The project was completed at a monetary cost of **BDT 0**. This was a design constraint rather than a happy accident: the project rule was free and open-source first, and paid service tiers were rejected whenever they were considered.

Table X itemises the resources used and the free provision that covered each.

TABLE X. Project budget for CSE499A and CSE499B.

Item	Source and justification	Cost (BDT)
Graphics-processor compute for the benchmarks	Google Colab free tier	0
Speech model checkpoints (18 models)	Public model hub, open licences	0
Audio corpus (≈4.7 h, 5 varieties)	Publicly available recordings, collected by the team	0
Project storage (≈5 GB)	Cloud drive free tier	0
Source-code hosting	Public repository free tier	0
Language-model services	Free tiers of four providers; usage on the metered account was verified at zero	0
Speech synthesis	Free keyless neural voice, with a fully local offline fallback	0
Short-message channel for one-time codes	Open-source gateway running on the team's existing subscriber module	0
Web search for the drug-information assistant	Free keyless search library	0
Software libraries and frameworks	All open-source	0
Development machines	The team's own existing computers; no purchase was made	0
Hosting, domain or cloud deployment	None. The system runs locally from a single command	0
Total cost		**0**

6.2.1 Costs a Real Deployment Would Carry

A zero-cost prototype is not a zero-cost product, and the distinction is worth making explicitly.

Three items in the table above are free only because this is a development project. The language-model providers are free tiers with daily quotas and terms that permit training on submitted data; a real clinic would need a paid provider that contractually does not train on inputs, or a local model. The short-message channel runs on the team's own subscriber module, which is acceptable for a demonstration but not for a clinic; the production route is a licensed national aggregator, priced per message. The speech recogniser is free because it is the browser's own cloud service, which is exactly the property that makes it unsuitable for real patient audio.

A deployment would also add hosting, a domain and a certificate, and the engineering time for the authentication and encryption work described in Section 8.3. None of these were incurred by this project, and none are estimated here, because an estimate not grounded in a quotation would be a guess.

Chapter 7 Complex Engineering Problems and Activities

This chapter maps the project against the complex engineering problem and complex engineering activity attributes required for accreditation. The knowledge-profile mappings reflect the software, machine-learning and signal-processing character of the work, and the two tables were prepared in consultation with the faculty advisor.

7.1 Complex Engineering Problems (CEP)

Table XI maps each of the seven complex engineering problem attributes onto the work carried out in this project.

TABLE XI. Complex Engineering Problem (CEP) attributes addressed by this project.

Attribute		How the project addresses it
P1	Depth of knowledge required (K3–K8)	The project required software engineering, data structures and relational database design (K3); specialist knowledge of machine learning, speech signal processing and natural language processing (K4); the design of an end-to-end multi-stage system with a stateful conversational loop, a versioned schema and a provider-failover layer (K5); competence with modern engineering and information technology tools including Python, a web framework, an object-relational mapper, migration tooling, model libraries, audio processing tools and a text shaping engine (K6); an understanding of the role of engineering in society, covering patient privacy, clinical safety and responsible use of artificial intelligence in medicine (K7); and engagement with current peer-reviewed literature on multilingual speech recognition, low-resource language processing and clinical language models (K8).
P2	Range of conflicting requirements	Several requirements pull against each other and had to be traded explicitly. Recognition accuracy conflicts with latency and with running on a machine that has no dedicated graphics processor. Patient privacy conflicts with the use of free cloud speech and language services, which is why development was restricted to synthetic data. Safety recall conflicts with precision: the danger-sign rule is deliberately tuned to over-trigger, accepting false alarms in exchange for not missing a real one. Automation conflicts with human authority: the system extracts fields automatically, but a staff correction must never be overwritten by a later automatic pass. Thoroughness of questioning conflicts with patient fatigue, which is why the follow-up loop has both a floor and a ceiling on the number of questions.

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Table XI continued from previous page

Attribute		How the project addresses it
P3	Depth of analysis required	There is no unique correct design. Selecting a speech recognition approach required a controlled empirical comparison of eighteen models under three metrics, followed by a reasoned choice that took accuracy, licence, hardware requirement and deployability together. The system architecture required comparable analysis: sixty-nine design decisions were recorded during the project with their alternatives and the reason for rejecting each, covering the database engine, the front-end approach, the speech and synthesis paths, the provider strategy, the identity mechanism and the document formats. Several decisions were later superseded on new evidence, which is itself recorded rather than hidden.
P4	Familiarity of issues	The team worked across several unfamiliar and infrequently encountered areas: the principal open speech model families and their differing interfaces and failure modes; audio preprocessing including resampling, loudness normalisation and voice-activity segmentation; standard speech evaluation metrics and their edge cases, including error rates above unity caused by hallucinated output; the behaviour of large multimodal audio models when given out-of-distribution input; complex-script text shaping, where a general document library silently drops Bengali conjuncts; and the practical behaviour of free service quotas, which fail in different ways depending on whether the limit is per model queue or per account.
P5	Extent of applicable codes	No single engineering code or standard governs a clinical pre-screening tool of this kind in Bangladesh. The project therefore adopted relevant conventions voluntarily: the HL7 FHIR R4 specification for the exported clinical record, so the output could in principle be consumed by a real records system; conventional software engineering practice including version control, versioned and reversible schema migrations and an automated regression suite; and professional codes of ethics for computing in the handling of patient data. It is stated openly that the interchange export is structurally valid and conservative but is not certified, claims conformance to no implementation guide, and has never been tested against a real receiving system. A deployed system would additionally have to comply with whatever national health-data framework applies at the time.
P6	Extent of stakeholder involvement	The system has several distinct stakeholder groups with different and sometimes competing interests. Patients are both the users of the kiosk and the subjects of the data, and include elderly and low-literacy people whose needs drove the voice-first design. Triage staff need to decide who is seen next and whether a case is fit to hand over. Physicians consume the record and carry the clinical responsibility, which is why the system never diagnoses. Clinic administrators would own any deployment and its cost. The department and the faculty advisor provided academic oversight. National health-data regulators would govern any real use. The design reflects these differences directly: the three portals are three roles with different data ownership, not three views of one screen.

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Table XI continued from previous page

Attribute		How the project addresses it
P7	Interdependence	The system is a chain of interdependent stages that must operate in series: speech capture, correction, extraction, summarisation, gap analysis, question generation, answer merging, completeness scoring, risk assessment, explanation, report assembly, storage, and the staff-facing review and export. An error at an early stage propagates to every later one, which is precisely the risk that the measured speech recognition error rate creates. The design responds with per-stage interfaces, an immutable record of the original input, staged rather than in-place transformation, safe defaults at every failure point, and a safety rule that is independent of the rest of the chain. Sub-systems that fail independently — speech recognition, speech synthesis, four model providers, the message channel — each have their own fallback.

7.2 Complex Engineering Activities (CEA)

Table XII maps the five complex engineering activity attributes onto the same work.

TABLE XII. Complex Engineering Activity (CEA) attributes addressed by this project.

Attribute		How the project addresses it
A1	Range of resources	The activity drew on human resources (a three-member team and the faculty advisor), a self-assembled multi-dialect audio corpus, openly licensed model checkpoints from a public model hub, free-tier cloud graphics-processor compute, free tiers of four language-model providers, a broad set of open-source libraries spanning web services, databases, migrations, documents, printable output, text shaping, speech synthesis and testing, peer-reviewed research literature, and the team's own ordinary computers, which have no dedicated graphics processor.
A2	Level of interactions	The work required continuous interaction within the team, structured interaction with the faculty advisor through periodic demonstrations and design reviews, and indirect interaction with the open-source community through model documentation and published evaluation practice. It also required resolving interactions between components that were not designed together: a browser speech interface, four independent model providers, a document library, a text shaping engine and a message gateway. A deployed system would extend this to clinics, physicians, patients and health-data regulators.

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Table XII continued from previous page

Attribute		How the project addresses it
A3	Innovation	The project applies modern speech and language technology to a setting that has not been benchmarked in the published literature: the simultaneous combination of Bangladeshi regional dialect, Bangla–English code-mixing and informal clinical description. To the best of the team's knowledge, the finding that large multimodal audio models of 1.7 to 7 billion parameters all perform worse than a smaller Bangla-specialised model on this material, with two failing to produce usable output, is not reported elsewhere. The system design is also a response to that measurement rather than a standard pipeline: preserving the original words permanently, transforming only in separately stored stages, recovering lost information by conversation, and keeping the emergency check as a local rule that no model can suppress.
A4	Consequences to society and the environment	The intended social consequence is better clinical documentation for patients currently served by none, and access for people excluded by literacy, age or dialect from English-first digital health tools. The intended safety consequence is that danger signs reach the doctor and are never explained away by a failing model. Against this, the project carries real risks that are stated rather than minimised: an inaccurate transcript could mislead if a clinician trusted it without reading the patient's own words, and the current absence of authentication and encryption makes the system unsuitable for real patient data. The environmental footprint is deliberately small: no model was trained, both benchmarks were inference-only, the delivered system runs on ordinary hardware, and the planned adaptation method trains a small fraction of a model's weights. The work contributes to United Nations Sustainable Development Goal 3, Good Health and Well-being, and Goal 10, Reduced Inequalities.
A5	Familiarity	The activity required familiarity well beyond a normal coursework project: deep learning frameworks and model libraries; the Whisper, wav2vec 2.0, massively multilingual and audio language model families; audio signal processing and voice-activity segmentation; low-resource language processing; clinical language model literature; relational schema design with versioned migrations; asynchronous web service design; browser speech recognition and synthesis interfaces and their error behaviour; complex-script typography; the HL7 FHIR R4 document structure; applied cryptographic practice for one-time codes including salted hashing, constant-time comparison and attempt limiting; and disciplined software engineering practice including regression testing, recorded design decisions and version control.

Chapter 8 Conclusions

8.1 Summary

This project set out to let a patient in a Bangladeshi clinic describe a medical problem by speaking naturally, in Bangla, in Banglish or in a regional dialect, and to turn that conversation into a structured record the doctor can read before the consultation starts.

The work began by measuring rather than assuming. A multi-dialect Bangla speech corpus of about 4.7 hours across five regional varieties was assembled and preprocessed, and eighteen open-source models were benchmarked on it. The best model, a Bengali regional fine-tune of Whisper Medium, reached a Word Error Rate of 0.4694, missing close to half the words. Six larger multimodal audio models were then tested on the same clips: all four that produced output scored above 1.0, and two produced nothing usable at all. Scale did not close the gap.

The second half of the project treated that result as a design constraint. The delivered system keeps the patient's exact words permanently and never edits them, performs cleaning and structuring in later and separately stored steps, recovers the information that recognition loses by asking spoken follow-up questions over ten fixed clinical fields, and keeps the emergency safety check as a deterministic local rule that no language model can suppress or override downward. Around that pipeline sit three portals for the patient, the triage desk and the doctor, an eighteen-table database managed by fourteen versioned migrations, a hashed one-time-code identity flow, an append-only audit trail, a multi-provider failover chain for the language services, and clinical output as editable documents, a standards-based interchange record and a correctly shaped Bangla printable file.

The behaviour of that system is protected by an automated suite of 89 test files whose last recorded run passed 1,196 tests with 2 skipped and no failures, and the complete voice loop has been demonstrated end to end once with a real microphone in a supervised session. What the project has *not* done is measure the built system: there is no accuracy, precision, latency or usability figure for it, and this report has been careful throughout not to borrow the benchmark's numbers to fill that gap.

8.2 Limitations

The following limitations are stated in full. Several of them are uncomfortable, and all of them are real.

Bangla dialect speech recognition remains the binding constraint. The best open model available still loses roughly half the words, and at a threshold of 0.3 Word Error Rate no model recovered a single Barishal clip cleanly. Every downstream stage inherits this.

The benchmark covers three of five varieties. Error rates were computed over 147 of 250 evaluation clips. Puran Dhaka and Sylheti contributed nothing, because their reference transcripts could not be paired with the processed audio files. The dialect-degradation finding therefore rests on Barishal alone.

Two large models were never scored. Phi-4 Multimodal and Qwen2.5-Omni produced no usable output, so the comparison of large models covers four, not six.

The deployed system has no measured evaluation. There is no Word Error Rate for the browser recogniser the demonstration uses, no extraction precision or recall, no risk-tier accuracy against clinical judgement, no latency distribution, and no usability study. The supervised live session is qualitative evidence that the loop works, not a measurement of how well.

Speech recognition and the language layer both depend on the cloud. Audio goes to a third-party recogniser and the language tasks go to external providers whose free tiers may train on submitted inputs. This is the direct reason development was confined to synthetic data.

There is no authentication and no encryption. Staff select a seeded account rather than logging in, and nothing is encrypted at rest or in transit. The system cannot hold real patient data in its present state.

Provider failover has only been tested against fakes. The multi-credential chain has never been exercised against real providers under real quota pressure.

One specified voice-loop step was not built. Silence re-prompting, an empty-submission guard, an answer-length cap and permission recovery were specified and deliberately left unbuilt, and no microphone session has been run since the last change to the speech code.

The feedback loop is half a loop. Doctor feedback is captured and stored, but there is no retraining or regression pipeline to consume it.

The interchange export is unproven in practice. It is structurally coherent and conservative, but it is not certified, claims conformance to no implementation guide, and has never been sent to a real records system.

The system has never met a patient. No clinical trial, no pilot, no usability testing with elderly or low-literacy users. Every benefit described in Chapter 5 is a designed-for benefit.

8.3 Future Improvement

The limitations above map fairly directly onto the work that should come next.

Complete the benchmark. Repairing the filename mismatch between the reference transcripts and the processed audio would recover Puran Dhaka and Sylheti and produce a genuine five-variety result. This is the cheapest improvement available and the one that would most strengthen the research contribution.

Evaluate the built system. A formal study on a labelled set of recorded consultations should measure Word Error Rate and Character Error Rate for the deployed recogniser, precision, recall and F-measure for each of the ten extracted fields, agreement of the risk tier with clinician judgement, and the latency distribution from speech to displayed text and from submission to a completed report. These are the metrics a future evaluation should report; none of them is a result of this project.

Move recognition on-device. Replacing the browser's cloud recogniser with a local model would remove the privacy objection that currently confines the system to synthetic data, and would let it work without connectivity. A processor-only quantised model is the practical route on the hardware a clinic is likely to own.

Adapt a model to the dialects. With the corpus already collected, parameter-efficient adaptation of a mid-sized Bangla model is a realistic next step and was the methodology studied but not executed in the first semester.

Build the security layer. Real staff authentication with role-based access control, transport encryption, encryption at rest, a consent-management interface and a retention policy are all prerequisites for touching real patient data. The user table and the consent flag already exist as the seams for this.

Close the feedback loop. Once a locally trained model exists, the stored doctor corrections become training signal, and a regression check before any model update becomes necessary rather than optional.

Finish the voice loop and re-test it. The unbuilt silence and recovery step should be completed, and a fresh supervised microphone session run afterwards, since none has been run since the speech code last changed.

Prepare for deployment. Moving from a single database file to a database server, packaging the application as a container, and then running a small supervised pilot in one clinic with ethical clearance would turn the designed-for benefits of Chapter 5 into something that can actually be measured.

The project's most useful outcome may be the least glamorous one. By measuring Bangla dialect speech recognition instead of assuming it, the work established that the transcript cannot be trusted, and then showed that a useful clinical system can still be built on top of an unreliable transcript, provided the design never pretends otherwise.

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